

# FIRE ECOLOGY





# Fire as an ecosystem process

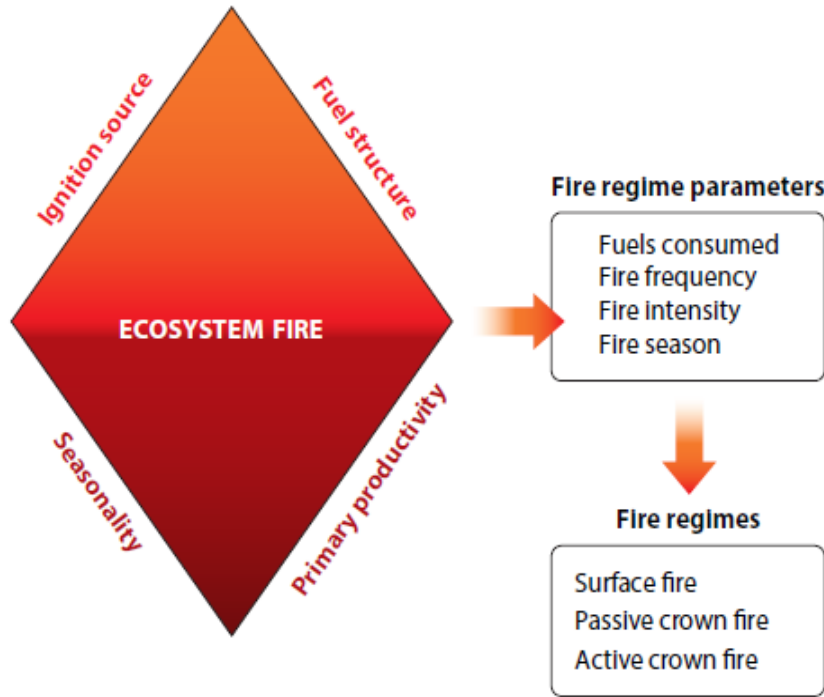


Figure 1

Factors necessary and sufficient for predicting the distribution of fire as an ecosystem process. A certain level of primary productivity is needed in order to spread fire, and climatic seasonality, with annual to decadal cycles of drying, is required to convert these potential fuels to available fuels that can combust. Fire regime parameters include the types of fuels consumed, fire frequency, fire intensity, and fire season.

Fire is an ecosystem process dependent on **plant community flammability** and enough **biomass continuity** for fire spread. In response, species evolve along different trajectories.

Key to understanding fire adaptations is the **fire-regime** concept, which involves **the conditions sufficient to explain the ecological patterns of fire activity** and includes four parameters:

- primary productivity,
- seasonality,
- ignitions,
- fuel structure.

**Mediterranean-type climate (MTC) shrublands** are very fire prone because they are characterized by a predictable annual drought of 3–4 months, in ecosystems with enough biomass to carry fire.



# FIRE ECOLOGY

## ***Mediterranean-type ecosystems***

***Geography***: west coasts of continents between 32° and 40° latitude

***Climate***: warm dry summers, cool moist winters, with a pronounced summer dry period

***Physiography***: landscapes of eroded mountains, heterogeneous low-nutrient soils, and pulse-driven river systems.

***Plant morphology and function***: communities dominated by ***evergreen sclerophyllous shrubs*** with many species displaying fire-cued seed germination

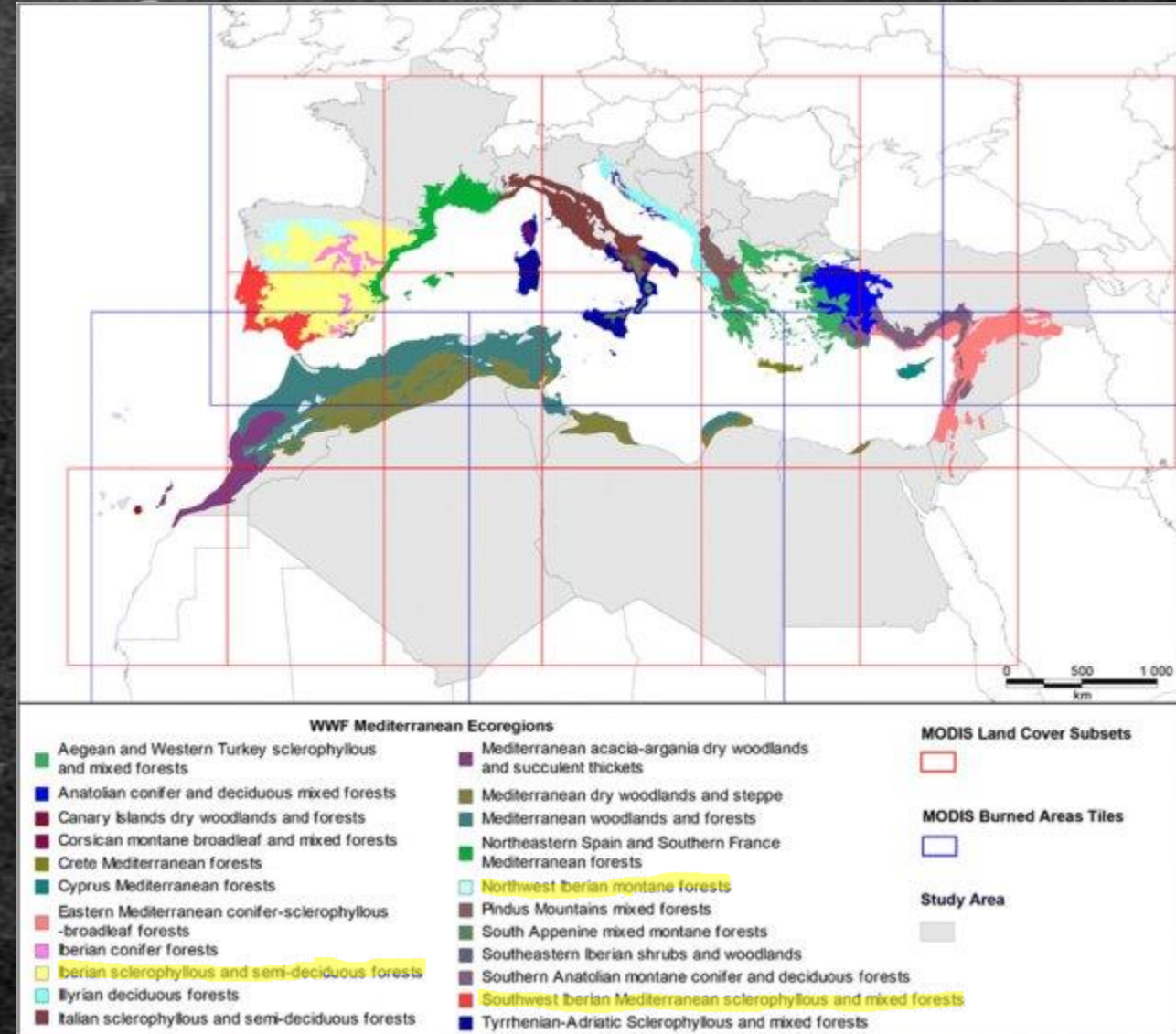


# FIRE ECOLOGY

Mediterranean biome



Ecoregions of the Mediterranean Basin



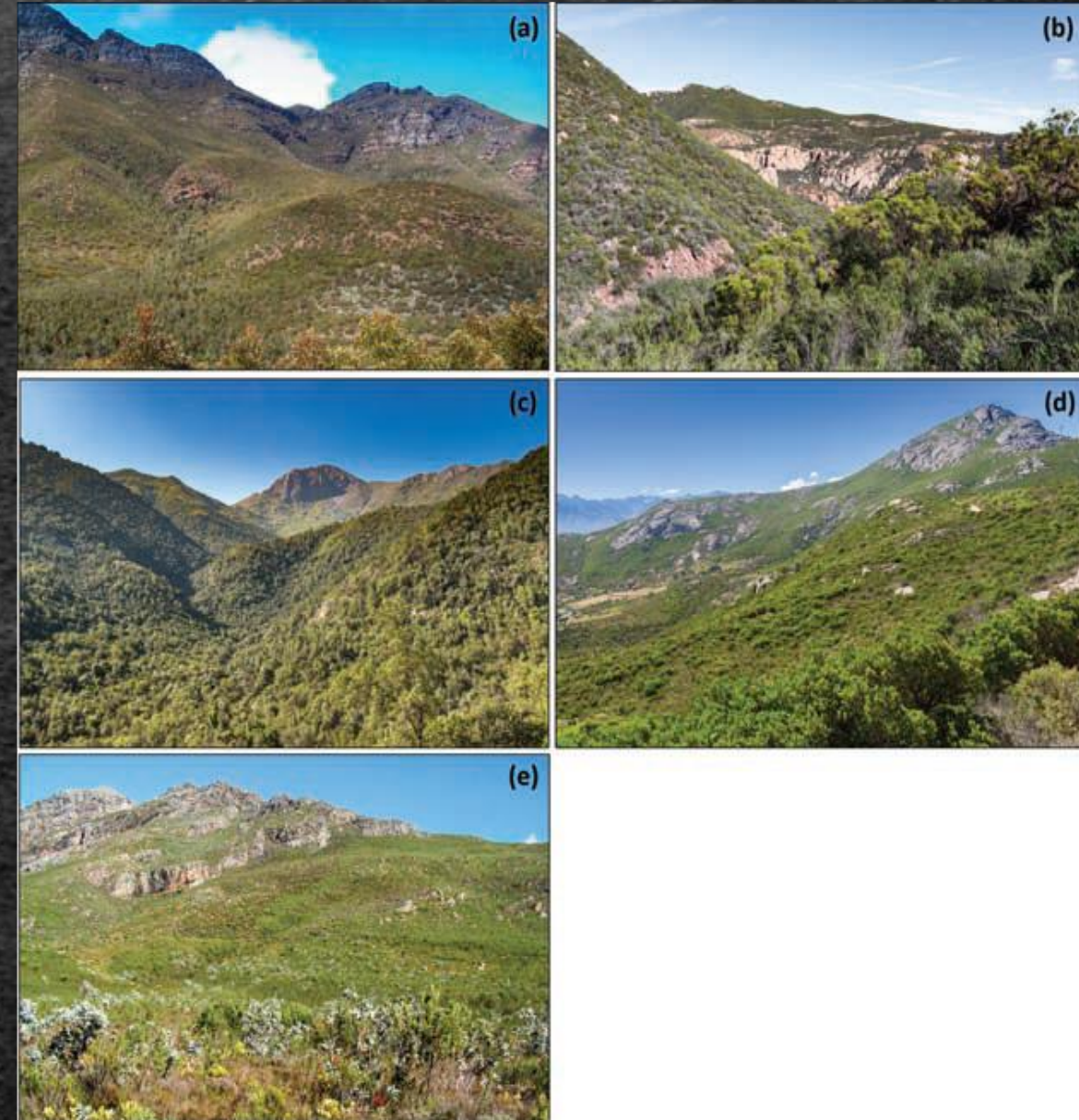


# FIRE ECOLOGY

The five regions are dominated by vegetation that forms a structurally similar **shrubland** where the dominant species are **evergreen** shrubs with **sclerophyllous** leaves.

Stirling Range, Western Australia, Australia, (b) Santa Ynez Mountains, California, USA, (c) Cordillera de la Costa, Valparaíso Region, Chile, **(d) Corsica, France**, and (e) Western Cape, South Africa.

Each region and country uses different names for this vegetation type and, for the shown images, the regional vegetation type names are: (a) mallee, (b) chaparral, (c) matorral, (d) maquis, and (e) fynbos.





# Fire adaptations

**Adaptive traits** are characteristics that **provide a fitness benefit**, and those traits vary depending on the fire regime.

Traits that are adapted to fire fall into two categories: those that **enhance survival** and those that **promote postfire regeneration**.

Understanding patterns of trait distribution requires consideration of the fire regime, as **traits adaptive under a crown-fire regime differ considerably from traits selected for under a surface-fire regime**; this is true for traits involved in both postfire survival and postfire regeneration.

Postfire survival is an **individual-level resistance** strategy, while postfire regeneration is a **population-level resilience** strategy.



# Fire adaptations

**Ecological resistance** is the ability of an ecosystem to **withstand or endure change** when faced with a disturbance.

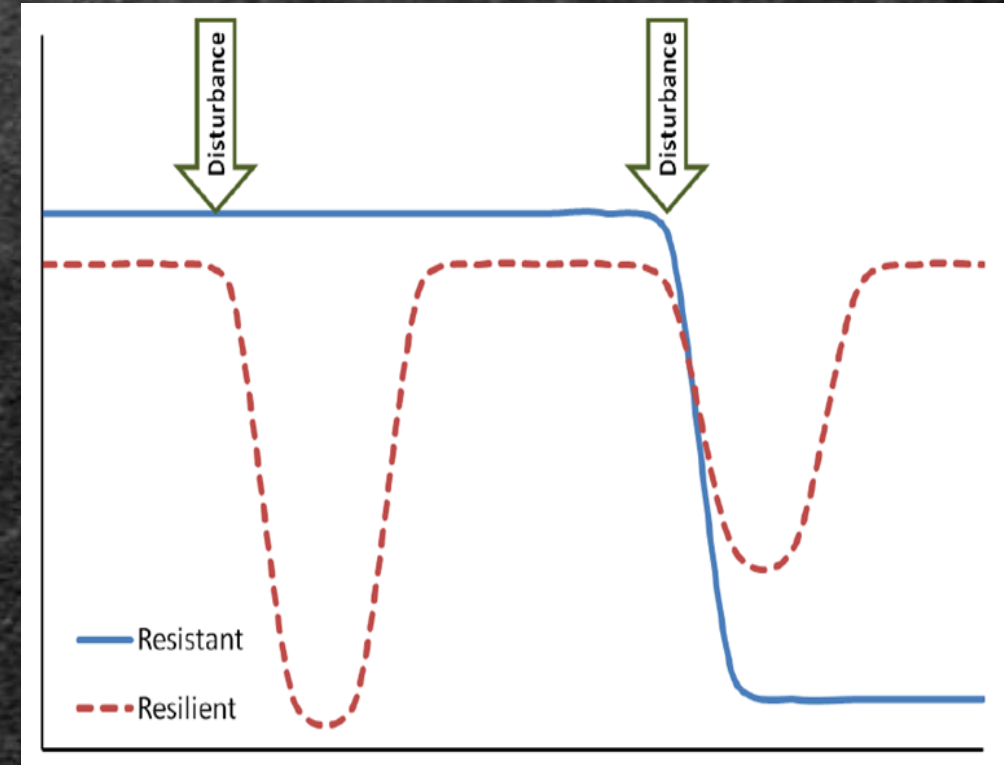
It measures the degree to which an ecosystem **maintains** its structure, function, and composition in the face of external pressures.

Highly resistant ecosystems are able to **maintain** their integrity and stability even when subjected to disturbances such as droughts, fires, or invasive species.

**Ecological resilience** is the ability of an ecosystem to **recover and return** to a stable state following a disturbance.

It measures the capacity of an ecosystem to **absorb and adapt to changes** while maintaining its essential structure, function, and identity.

Resilient ecosystems are able to **bounce back** from disturbances and continue to provide ecosystem services and support biodiversity.





# FIRE ECOLOGY

## **Postfire regeneration traits**

- **Postfire resprouting**: the ability to generate new shoots from dormant buds after stems have been fully scorched by fire. Species are typically classified as resprouters or nonresprouters depending on this ability. There are different types of resprouting depending on the location of the buds (epicormic, lignotuber, rhizome, roots, etc.).
- **Postfire seeding**: the ability to generate a fire-resistant seed bank with seeds that germinate profusely after fires (fire-cued germination). Typically, such species restrict recruitment to a single pulse after a fire. Seeds may be stored in the soil or in the canopy. Species are typically classified as seeders or nonseeders (or fire-dependent/fire-independent recruiters) depending on this ability. There are different types of postfire seeding.



# FIRE ECOLOGY

## Postfire strategies

- **Obligate resprouters**: plants that **rely on vegetative resprouting** to regenerate after fire (resprouters without postfire seeding ability). These plants **do not germinate after fire because they lack a fire-resistant seed bank**. Note that obligate resprouters might reproduce by seeds during the fire-free interval, but the terminology of seeders and resprouters refers to the postfire conditions. These are **iteroparous species** and have a **polypyric life cycle**.
- **Facultative seeders**: plants that **have both mechanisms for regenerating after fire**, that is, they are able to resprout and to germinate after fire.
- **Obligate seeders**: plants that **do not resprout and rely on seeding to regenerate their population after fire** (nonresprouters with postfire seeding ability). Because they tend to recruit massively once in their lifespan (after fire) and fire kills the adults and typically exhausts their seed bank, they can be considered **semelparous species** with nonoverlapping generations and a **monopyric life cycle**. Note that the term 'seeders' refers strictly to postfire conditions, and cannot be attributed to plants that regenerate by seeds in other conditions.
- **Postfire colonizers**: plants that lack a mechanism for local postfire persistence, but they recruit after fire by seeds dispersed from unburned patches or from populations outside the fire perimeter (metapopulation dynamics).



# FIRE ECOLOGY

Obligate  
resprouters

## *Quercus coccifera*

CARRASCO, CARRASQUEIRO

**Ecologia:** matagais; em solos pedregosos, ácidos ou básicos.



## *Erica australis*

URGUEIRA, URZE-VERMELHA, TORGA-VERMELHA

**Ecologia:** matos acidófilos; em locais soalheiros.



## *Arbutus unedo*

MEDRONHEIRO, ÊRVODO, ERVEDEIRO

**Ecologia:** matagais, bosques perenifólios; indiferente edáfica.



Facultative  
seeder/  
resprouter

## *Pterospartum tridentatum*

CARQUEJA, CARQUEIJA

**Ecologia:** matos; em solos ácidos; do litoral à alta montanha.



## *Cytisus multiflorus*

GIESTA-BRANCA

**Ecologia:** matos, pastagens; em solos ácidos e pobres.



Obligate  
seeders

## *Cistus ladanifer* subsp. *ladanifer*

ESTEVA, XARA

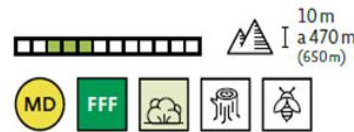
**Ecologia:** matos; em solos secos e pobres, ácidos.



## *Cistus albidus*

ROSELHA-GRANDE

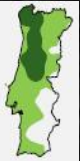
**Ecologia:** matos; em solos secos, geralmente básicos.



## *Erica umbellata*

QUEIROGA, QUEIRÔ, TORGA

**Ecologia:** matos acidófilos, pinhais.





# FIRE ECOLOGY

## **Fire-stimulated germination**

**Fire promotes the germination of soil-stored seeds.** This ensures that germination is cued to post-fire conditions when light, water and nutrient resources are optimal.

Hard seeds increase fitness via three unique, interacting mechanisms:

- (i) they **increase seed longevity** that may sometimes reach over 100 years
- (ii) they **enable seeds to germinate quickly in response to fire heat** that signals an improvement in conditions suitable for seedling establishment
- (iii) they ensure survival in the digestive tract of (occasional) dispersal agents

The *Fabaceae* (*Leguminosae*) are the most important family with heat-stimulated germination of hard seeds.

Recurrent fire will promote strong hard-seededness (i.e. requires fire-associated temperatures to break dormancy), as optimal conditions for recruitment may exceed decade-long intervals.



# Fire adaptations

Plant responses to fire may be characterized by whether they **resprout (R)** or not and have **recruitment stimulated by fire (S)** or not, in four possible combinations:

- Resprouters with fire stimulated recruitment (R+S+). **Facultative seeder / resprouter** → *some plants die, seeds germinate*.
- Resprouters without fire stimulated recruitment, (R+S-). **Obligate resprouters** → *few plants die, underground biomass unburned*.
- Non-resprouters with fire stimulated recruitment, (R-S+). **Obligate seeders** → *all plants die, seeds germinate*.
- Non-resprouters without fire-stimulated recruitment (R-S-). **Fire avoiders**, *all plants die, all biomass is lost* → *no adaptations to fire*,

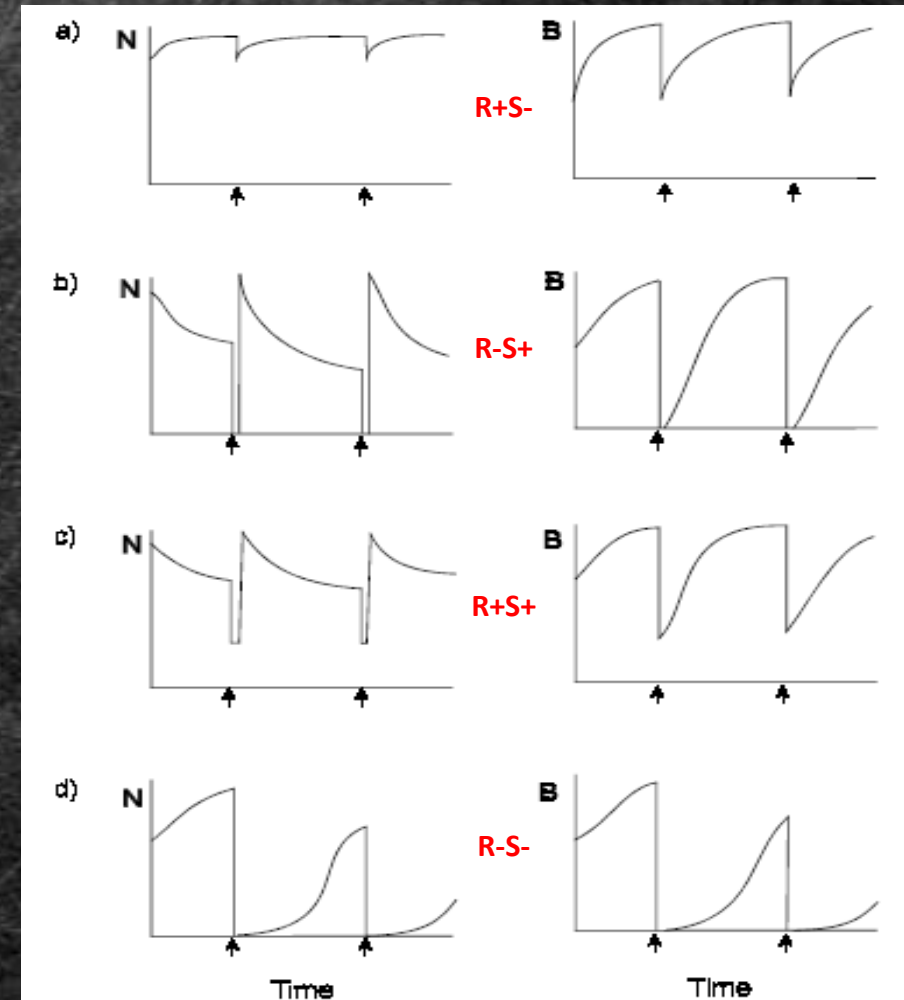


Figure 2. Possible population dynamic trends for different functional types (N: population size; B: biomass of the population, including below-ground biomass but not seed biomass). Maximum N and B are scaled for each functional type. Arrows indicate fires. (a) resprouting species; (b) species unable to resprout after fire but with recruitment stimulated by fire (disturbance-dependent recruitment); (c) resprouting species with the recruitment stimulated by fire and (d) non-resprouters with recruitment not-stimulated by fire



# Fire adaptations

*Table 3.* Relation between different life history traits and the four functional types in relation to the regenerative response after fire. **The four functional types are defined by the combination of the first two life-history traits** (in bold).

| Life-history traits                 | <b>R+S-</b> Functional types                                                                                                                                                                                      | <b>R+S+</b>                                                                                                                                                                                                          | <b>R-S+</b>                                                                                                              | <b>R-S-</b>                                                                                                                                  |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Resprouting ability                 | yes (high)                                                                                                                                                                                                        | yes (intermed.)                                                                                                                                                                                                      | no                                                                                                                       | no                                                                                                                                           |
| Fire-stimulated recruitment         | no                                                                                                                                                                                                                | yes                                                                                                                                                                                                                  | yes                                                                                                                      | no                                                                                                                                           |
| Life span                           | long                                                                                                                                                                                                              | long/intermed                                                                                                                                                                                                        | short                                                                                                                    | short                                                                                                                                        |
| Growth rate                         | low                                                                                                                                                                                                               | interm.                                                                                                                                                                                                              | high                                                                                                                     | high/interm.                                                                                                                                 |
| Root system                         | deep                                                                                                                                                                                                              | intermed.                                                                                                                                                                                                            | intermed/shallow                                                                                                         | shallow                                                                                                                                      |
| Root/shoot biomass ratio            | large                                                                                                                                                                                                             | large/interm.                                                                                                                                                                                                        | low                                                                                                                      | low                                                                                                                                          |
| Type of dispersal units             | big, fleshy or<br>acombs,<br>nonrefractory                                                                                                                                                                        | small, light,                                                                                                                                                                                                        | small, light, hard,<br>refractory                                                                                        | small, light,<br>nonrefractory                                                                                                               |
| Number of dispersal units           | few                                                                                                                                                                                                               | intermed.                                                                                                                                                                                                            | many                                                                                                                     | many                                                                                                                                         |
| Dispersal vector                    | animals                                                                                                                                                                                                           | variable                                                                                                                                                                                                             | wind                                                                                                                     | wind                                                                                                                                         |
| Seed bank                           | no                                                                                                                                                                                                                | variable                                                                                                                                                                                                             | yes (soil or<br>canopy)                                                                                                  | variable?                                                                                                                                    |
| Seed life span                      | short                                                                                                                                                                                                             | short/variable                                                                                                                                                                                                       | long                                                                                                                     | short                                                                                                                                        |
| Seedling establishment              | low                                                                                                                                                                                                               | intermediate                                                                                                                                                                                                         | high                                                                                                                     | high                                                                                                                                         |
| Leaf mass/area ratio                | high                                                                                                                                                                                                              | variable                                                                                                                                                                                                             | low                                                                                                                      | low                                                                                                                                          |
| Shade tolerance                     | high                                                                                                                                                                                                              | low                                                                                                                                                                                                                  | low                                                                                                                      | low                                                                                                                                          |
| Drought tolerance                   | low                                                                                                                                                                                                               | high                                                                                                                                                                                                                 | high                                                                                                                     | variable                                                                                                                                     |
| Susceptibility to disturbance       | low                                                                                                                                                                                                               | intermediate                                                                                                                                                                                                         | high                                                                                                                     | high                                                                                                                                         |
| Susceptibility to climate<br>change | high                                                                                                                                                                                                              | intermediate                                                                                                                                                                                                         | low                                                                                                                      | low                                                                                                                                          |
| Fire-related mortality              | low                                                                                                                                                                                                               | low/intermed.                                                                                                                                                                                                        | high                                                                                                                     | high                                                                                                                                         |
| Examples                            | <i>Quercus ilex</i><br><i>Q. coccifera</i><br><i>Q. calliprinos</i><br><i>Arbutus unedo</i><br><i>Rhamnus sp. pl.</i><br><i>Phyllirea sp. pl.</i><br><i>Pistacia lentiscus</i><br><i>Brachypodium<br/>retusum</i> | <i>Thymus vulgaris</i><br><i>Anthyllis cytisoides</i><br><i>Psoralea bituminosa</i><br><i>Genista scorpius</i><br><i>Dorichium</i><br><i>pentaphyllum</i><br><i>Oryzopsis miliacea</i><br><i>Eucalyptus globulus</i> | <i>Cistus sp. pl.</i><br><i>Pinus halepensis</i><br><i>P. brutia</i><br><i>Ulex parviflorus</i><br><i>Pinus pinaster</i> | <i>Taraxacum sp. pl.</i><br><i>Galactites tomentosa</i> ,<br><i>Conyza sp. pl.</i><br><i>Chenopodium album</i><br><i>Juniperus phoenicea</i> |



# Fire adaptations

## TERMINOLOGY

**Crown fires:** fires in woody-dominated vegetation that affect most of the crown of the dominant plants.

**Grass-stage:** delayed trunk elongation, whereby internode elongation is suppressed for the first 5–10 y of development, with the resulting juveniles persisting in a bunchgrass growth form. During this stage, needles are decumbent, accumulate as a mantle over the apex, and insulate the apical bud against fire heat.

**Fire regime:** the characteristics of wildfire activity that prevail in a given area, determined by its frequency, intensity, seasonality, and type of fuels consumed. Two contrasted fire regimes are surface-fire regimes and crown-fire regimes.

**Monopyric life cycle:** species that perform all their life cycle within a fire cycle. In plants, examples are annual and biennial species, and are post-fire obligate seeders (including the fire-embracers pine syndrome).

**Polypyric life cycle:** species that perform all their life cycle through multiple fire cycles. In plants, examples are those with fire survival strategies such as resprouting species or trees with thick bark (e.g., the fire-tolerator pine syndrome).

**Self-pruning** (= branch shedding): ability to shed lower branches and, thus, generate a fuel discontinuity between the understory and the canopy. The opposite is branch retention.

**Serotiny:** retention of matured seed in closed cones for more than a year in such a way that seed dispersal is delayed. The heat of the fire acts as a cue to open the cones; thus, seeds are dispersed post fire, when conditions for establishment are favorable (resources, such as light, moisture, and nutrients, are maximal and competition with adults is minimal).

**Understory fires:** surface fire in forest ecosystems; that is, fires that spread through the herbaceous and litter layer (surface fuels) of forests, without affecting the tree crowns. Forests subject to understory fires typically have a vertical discontinuity of fuels (e.g., lack of lower branches in the bole) in such a way that tree crowns are not affected by the fire.

**Semelparous species:** those characterized by a single reproductive episode before death.

**Iteroparous species:** those characterized by multiple reproductive cycles over the course of their lifetime.



# Fire adaptations promoting postfire survival

**Resprouting** from surviving vegetative structures is the primary means of persistence after fire.

This commonly occurs from **preformed buds at the base of stems** (in the root crown) or in **belowground organs** (basal resprouting). Belowground organs with a resprouting bud bank in woody species include **roots, lignotubers, and woody rhizomes**. Herbaceous perennial species resprout from **bulbs, corms, tubers, or rhizomes**.

A specialized form of resprouting common in Mediterranean ecosystems involves the **lignotuber**. It is a basal swelling formed during the early development of saplings and contributes to successful resprouting by **storing carbohydrates** and **dormant buds**.



(a) Resprouting from lignotuber in *Adenostoma fasciculatum* (chaparral, southern California).



# Fire adaptations promoting postfire survival

A different type of regeneration is **epicormic resprouting** (i.e., resprouting from **canopy stems**). It is common in Australian **Eucalyptus**, including **E. globulus**, widely planted in Portugal, and in Northern Hemisphere **Quercus**, namely the cork oak, **Q. suber**.

Evolution of resprouting is difficult to attribute to a response to fire, because it is so widespread in woody angiosperms. The problem in pointing to fire as a driver of resprouting is that it is also of adaptive value following other disturbances, such as extreme winds, severe frosts, drought, and animal browsing.

However, some specific resprouting structures, like **lignotubers** and **woody rhizomes**, occur **only in fire-prone ecosystems** and **likely evolved in response to fire**.



Cork oak (*Q. suber*) population two months after fire with characteristic epicormic resprouting.  
<https://jgpausas.blogs.uv.es/2018/10/15/el-surar-de-pinet-a-small-isolated-population-of-cork-oak/>



# Fire adaptations promoting postfire survival

**Thick bark** is another trait associated with survival, and it is common in woodland and savanna trees. Thick bark also occurs in *Pinus*, where it originated approximately 125 Mya, in the fiery environment of the Cretaceous\*.

In addition to traits that enhance postfire survival, some species persist in the face of severe fires by occupying **refugia**, such as humid riparian areas, or sparsely vegetated rock outcrops, which may escape destruction. These refugia provide **seed sources for reestablishing in burned areas**.



Cork, the bark of *Q. suber*, the cork oak. It can be a few cm thick and is a very good thermal insulator.

\* On average, throughout the Cretaceous (145 – 66 Mya),  $[O_2]$  was  $\approx 30\%$ , i.e. 50% higher than in the present.



# Fire adaptations promoting postfire recruitment

Some plant species are **obligate seeders**, i.e., they have no capacity for resprouting and are completely dependent on **seedling recruitment after fire**. There are also **facultative seeders** that **both resprout and recruit seedlings after fire**.

Plants display three **modes of recruitment after fire**:

- endogenous reproduction from dormant **seed banks stored in the soil**,
- endogenous reproduction from **canopy-stored seeds in serotinous cones or fruits**, and
- seedling recruitment from postfire **dispersal of seeds** produced by **surviving plants** or **unburned populations**.





# Fire adaptations promoting postfire recruitment

## RECRUITMENT FROM DORMANT SOIL-STORED SEED BANKS

Many species in fire-prone environments produce seeds that are **deeply dormant at dispersal** and **remain viable for decades to centuries**. Typically, **germination occurs in the first postfire growing season**.

Annual plants largely restricted to recruitment in the first postfire year are called **pyroendemics** or **fire ephemerals**. For many shrubs in Mediterranean-type ecosystems, fire produces a pulse of recruitment, resulting in an even-aged cohort of seedlings.

**Seed dormancy may be broken by heat shock or smoke**. **Heat-stimulated germination** is very common in Mediterranean-type ecosystems, namely in plant families *Anacardiaceae* (e.g. *Pistacia lentiscus*), *Cistaceae* (e.g. *Cistus* sp.), *Fabaceae* (*Ulex* sp., *Genista* sp.), and *Rhamnaceae* (*Rhamnus alaternus*).

In MTE, many species (e.g. *Erica* sp.) have deeply dormant seeds that require a chemical from biomass combustion, referred to as **smoke-stimulated germination**, although charred wood and other combustion products may also trigger germination.





# Fire adaptations promoting postfire recruitment

## RECRUITMENT FROM DORMANT CANOPY-STORED SEED BANKS

Recruitment may also occur from **serotiny** (from the Latin *serus*, meaning *late*), in which seeds are released from closed cones or fruits, with **opening normally delayed until after fires**.

Dormancy is the result of being enclosed in fruiting structures, and seeds germinate readily following postfire dispersal. **Serotiny is a fire-adapted trait found primarily in crown-fire ecosystems**, including shrublands and forests. Cones or fruits open *en masse* and disperse seeds within days of fire, producing a pulse of seedling recruitment that leads to even-aged cohorts, and further recruitment is limited until the next fire.

One limitation to serotiny is that when fire intervals are very long and plants die before the next fire, seeds are released but seldom recruit successfully in the unburned vegetation. Therefore, **canopy seed banks are less resilient to long fire intervals than are soil seed banks**, which can remain viable for many decades after the plant dies. **In the Mediterranean Basin serotiny is restricted to the conifers *Cupressus* and *Pinus***, which always occur in crown-fire regimes with limited tree survival following fire.





# Fire adaptations promoting postfire recruitment

## RECRUITMENT FROM POSTFIRE DISPERSAL

Some species recruit into recently burned sites by dispersal. Many forest types with moderate to high productivity have trees that **survive fire by outgrowing surface fuels and dropping dead branches to produce a gap between surface fires and their canopies**. However, they lack a dormant seed bank and are not serotinous.

Postfire regeneration occurs in subsequent years from seeds produced by surviving trees that take advantage of the postfire open spaces in the understory. This mode of regeneration is **typical of many trees found in surface-fire regimes**, such as temperate mixed conifer forests and is exemplified in the Mediterranean Basin by *Pinus nigra*.

This mode of recruitment, however, is **greatly affected by the size of gaps in the canopy**. In recent years, with increasing fuel levels due to fire suppression, as well as other global changes, **gap size has increased, and this can slow the rate of recruitment**.



# Adaptations associated with other fire-related traits

A common trait in many fire-prone ecosystems is **fire-stimulated flowering**, that is, profuse flowering after fire, with limited flowering between fires. It occurs in several MTE monocotyledon and dicotyledon geophytes (e.g., *Orchidaceae*, *Liliaceae*, *Iridaceae*, *Droseraceae*). It apparently **occurs in response to higher light levels, temperatures, and nutrient levels**.

**Flammability** also varies with fire regime, due to characteristics of **plant structure**, including **retention of dead branches** and **chemical composition**. Flammability may enhance **inclusive fitness**\* if the resulting fires kill neighboring, less flammable individuals, which increases the recruitment possibilities for the fire-adapted flammable individual. Flammability may have evolved as a form of **niche construction**\*.

In crown-fire ecosystems, **some plants increase reproductive success by enhancing their flammability**, i.e., their ignitability and fire intensity (**hot-flammable strategy**); this occurs in plants with fire-stimulated germination (postfire seeders) when fires open vegetation gaps that are safe sites for seedling recruitment.

\* Inclusive fitness is the number of offspring equivalents that an individual rears, rescues or otherwise supports through its behavior.

\* Niche construction is the process by which an organism alters its own (or another species') local environment.



# Adaptations associated with other fire-related traits

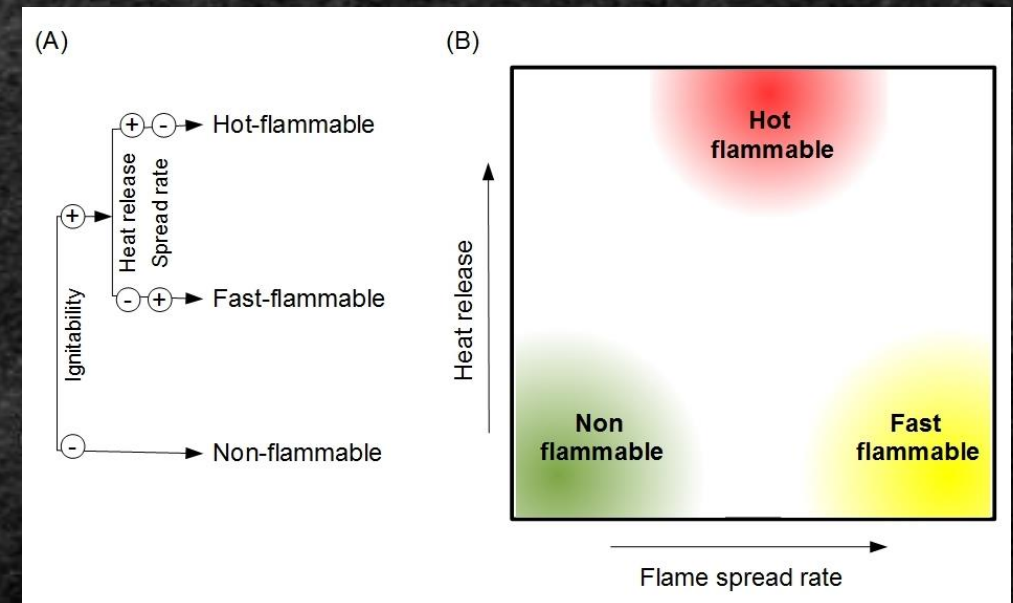
Examples of this hot-flammable strategy include:

**MTE shrubs** with high levels of standing dead biomass, thin twigs and leaves with high levels of volatile compounds, or low live-fuel moisture during the fire season,

**Pines that grow quickly and self-prune** their lower branches to avoid fire reaching the crowns,

Plants with a **highly flammable part (typically leaves) to ensure a quick burning (fast-flammable)** strategy, while protecting the other key parts of the plant where meristematic tissues are located.

These strategies represent alternative ways to persist in regimes with recurrent fires.





# Mediterranean pines and fire

The importance of fire as a selective agent in *Pinus* is revealed by the vast majority of taxa occurring in fire-prone environments.

Pine species that have thrived in fire-prone habitats, such as those found in Mediterranean climate zones, have most likely **developed traits to cope with fire**.

Fire-related traits of Mediterranean pines include **height, self-pruning of dead branches, bark thickness, seedling regeneration, needle characteristics, cone serotiny, and resprouting ability**.

Different combinations of these traits form **four distinctive syndromes** that have been specifically defined for pines (subgenus *Pinus*), probably the set of species most clearly shaped by evolution with fire:

- - Fire tolerator
- - Fire embracer
- - Fire avoider
- - Fire refugia



# Fire syndromes in pines

**Fire-tolerator** pines **survive frequent understory fires** thanks to their **thick bark** and **self-pruning** of the lower branches, which creates a gap between surface fuels and the canopy, with surviving trees providing a seed source for recruitment.

Fire-tolerator pines have **protected buds** and survive frequent, mostly understory fires.

*P. nigra* and *P. pinaster* (natural forests) are also fire-tolerators.



*P. pinea* (Umbrella pine, or stone pine) forest



# Fire syndromes in pines

**Fire-embracer** pines occur in **crown-fire regimes**. They often have **slow growth** rates that leave the canopy unable to outgrow surface fuels and a **hot-flammable strategy** that selects for the **retention of dead branches** and **serotiny**.

**Fire-embracers are obligate seeders**, retain their lower dead branches and empty cones, and are killed by crown fires. Their populations recover quickly after fire by massive germination of seeds from a **canopy-stored serotinous seed bank**.

Early reproduction is advantageous for serotinous species for avoiding **immaturity risk**, namely, being burned before the production of a sufficient large seed bank for eventual post-fire recruitment.



*P. halepensis* (Aleppo pine) forest

*P. brutia* is another fire-embracer species.



# Fire syndromes in pines

**Fire-avoider pines** occupy landscapes with **low fire incidence**, often in desert and alpine environments, and therefore lack fire adaptive traits.



*P. uncinata* forest

*P. sylvestris* is also a fire-avoider

**Fire refugia** pine species live in fire-prone ecosystems but **survive in parts of the landscape with limited fuel loads** (e.g., outcrops, riparian zones, grasslands).

Various species may display this syndrome, depending on the fire regime and topographic context.



Unburned and scorched pines in riparian / topographic refuge inside burned area.



# Fire syndromes in pines

Table 1. **Main fire traits** that define the three fire syndromes in pines.

| Fire trait <sup>b</sup> | Syndrome                                    |                                                  |                                               |
|-------------------------|---------------------------------------------|--------------------------------------------------|-----------------------------------------------|
|                         | Fire tolerator                              | Fire embracer (post-fire seeder)                 | Fire avoider                                  |
| Serotiny                | No                                          | Yes                                              | No                                            |
| Early reproduction      | No                                          | Yes                                              | No                                            |
| Thick basal bark        | Yes                                         | No                                               | No                                            |
| Resprouting             | No/Juveniles                                | No/Epicormic                                     | No                                            |
| Seedling grass stage    | (Yes)                                       | No                                               | No                                            |
| Self-pruning            | Yes (low tree flammability)                 | No (branch retention; high tree flammability)    | ~                                             |
| Litter flammability     | High (long, thin needles)                   | Low                                              | Low                                           |
| Examples                | <i>Pinus ponderosa</i> , <i>Pinus nigra</i> | <i>Pinus attenuata</i> , <i>Pinus halepensis</i> | <i>Pinus albicaulis</i> , <i>Pinus cembra</i> |

NOTES: The table does not consider the fire refugia syndrome because it does not depend on traits that enhance postfire plant survival.

*P. pinaster* is difficult to allocate to a single syndrome, due to variation among populations subject to different fire regimes. It may be a tolerator or an embracer.



# Fire syndromes in pines

Table 2. **Dominant characteristics of the fire and population dynamics** for each of the three main pine syndromes.

| Characteristic              | Syndrome                              |                                         |                              |
|-----------------------------|---------------------------------------|-----------------------------------------|------------------------------|
|                             | Fire tolerator                        | Fire embracer                           | Fire avoider                 |
| Fire type                   | Understory fires (localized crowning) | Crown fires                             | Infrequent fires             |
| Fire intensity              | Low                                   | High                                    | No, variable                 |
| Post-fire population status | Unaffected                            | Recruiting                              | Local extinction             |
| Persistence level           | Individuals                           | Populations                             | None (metapopulation)        |
| Generations                 | Long, overlapping                     | Short, nonoverlapping                   | Long, overlapping            |
| Life cycle [10]             | Polypyric                             | Monopyric                               | Fire independent             |
| Reproduction cycle          | Iteroparous                           | Semelparous                             | Iteroparous                  |
| Recruitment                 | Small gaps, fire dependent            | Large gaps, synchronous, fire dependent | Small gaps, fire independent |

NOTE: The table does not consider the fire refugia syndrome because it does not depend on traits that enhance postfire plant survival.



# Fire syndromes in Med Basin pines

**Table 22.1** Major Mediterranean pines, their habitats, risk of fire in their natural habitats (*in dense plantations there is a high risk of canopy fire for all species*) and their post-fire regeneration mode. *NA- North Africa, AC- Atlantic coast.*

| Pine species                                                                                              | Habitat                                  | Fire risk in nature                                              | Fire adaptation syndrome                                                                          | Post-fire regeneration                                                    |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <i>P. halepensis</i><br><i>P. brutia</i><br><b>Embracer</b>                                               | Low altitudes with Mediterranean climate | Extremely high risk of canopy fire                               | Thin-barked, killed by fire highly serotinous <sup>1</sup>                                        | Obligate seeders, massive post-fire seedling recruitment <sup>1</sup>     |
| <i>P. pinaster</i><br><b>Embracer</b> in NA and AC plantations.<br><b>Tolerator</b> in natural AC forests | Low altitudes with Mediterranean climate | Mostly surface fire; canopy fire in dense stands and plantations | NA: thin-barked, highly serotinous<br>AC: thick-barked, weakly serotinous, survives surface fires | NA: unknown<br>AC: recruitment is relatively good even after canopy fires |
| <i>P. pinea</i><br><b>Tolerator</b>                                                                       | Low altitudes with Mediterranean climate | Mostly surface fire; canopy fire in dense stands and plantations | Thick-barked, non-serotinous, self-pruning, survives surface fires                                | Recruits seedlings even after canopy fires                                |

|                                                                                |                                                                      |                                      |                                                                                                  |                                                                           |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <i>P. nigra</i><br><b>Tolerator</b>                                            | High mountains in the north, cold winter, cool summer with some rain | Rare fires in natural populations    | Thick-barked, non-serotinous, survives low severity, frequent surface fires, but not crown fires | Seed dispersal from unburned trees; low seedling recruitment              |
| <i>P. sylvestris</i><br><b>Avoider</b>                                         | High mountains, cold winter, cool summer with some rain              | Rare fires in natural populations    | Thin-barked, non-serotinous, does not survive crown fires                                        | Seed dispersal from unburned trees; low seedling recruitment              |
| <i>P. canariensis</i><br>Ambiguous                                             | Mid altitudes with Mediterranean climate                             | Frequent lighting and volcanic fires | Thick-barked, serotinous                                                                         | Adults have strong epicormic resprouting; no massive seedling recruitment |
| <i>P. heldreichii</i><br><i>P. mugu</i><br><i>P. unicata</i><br><b>Avoider</b> | High mountains, cold winter, cool summer with some rain              | Very rare fires                      | Thin-barked, non-serotinous, does not survive crown fires                                        | Seed dispersal from unburned trees; low seedling recruitment              |

<sup>1</sup>*Pinus halepensis* produces more cones, has a higher percentage of serotiny and recruits denser seedlings after canopy fires



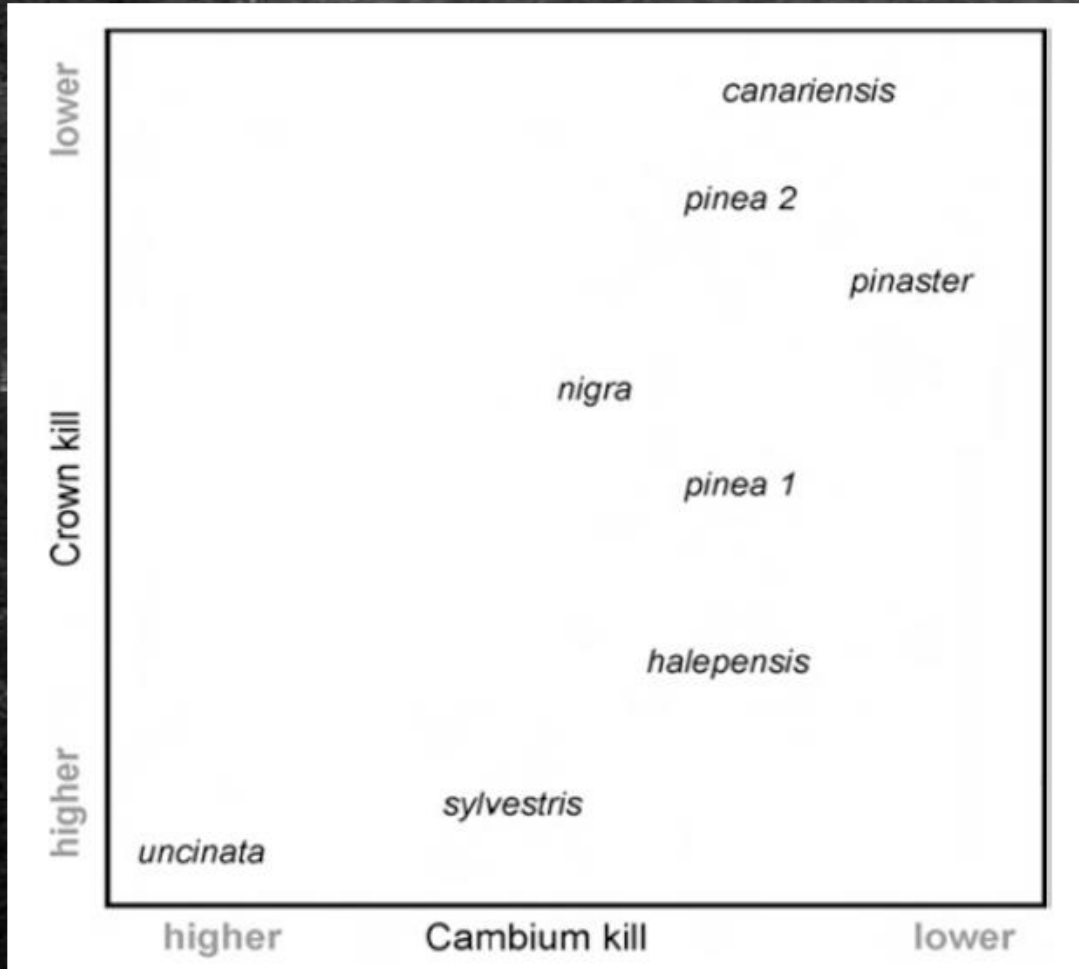
# Fire syndromes in Med Basin pines

*Table 4.* Life-history traits related to disturbances in the seven Spanish pines. \* Ruiz de la Torre 1979, (cm); ^Peraza 1964 (en Kg/L); \*\*Catalán 1991; "Krugman and Jenkinson 1974. (Sp: spring, Su: summer; A: autumn); # (Martínez-Millan et al. 1993, mean DBH of studied trees in parenthesis).

|                                                 | <i>Pinus</i>    |                   |              |                 |              |                   |                    |
|-------------------------------------------------|-----------------|-------------------|--------------|-----------------|--------------|-------------------|--------------------|
|                                                 | <i>uncinata</i> | <i>sylvestris</i> | <i>nigra</i> | <i>pinaster</i> | <i>pinea</i> | <i>halepensis</i> | <i>canariensis</i> |
| BUD: bud width (mm)                             | 5.0             | 5.5               | 8.5          | 8.2             | 6.5          | 5.5               | 11.5               |
| NEEDLE: needle length (cm)                      | 5.0             | 4.5               | 11.5         | 16.7            | 15.0         | 9.0               | 25.0               |
| WOOD: wood density (kg L <sup>-1</sup> )        | 0,502           | 0,522             | 0,598        | 0,475           | 0,596        | 0,548             | 0,753              |
| HEIGHT: maximum mature height (m)*              | 25              | 40                | 50           | 40              | 30           | 22                | 60                 |
| LONG: longevity (years)*                        | 600             | 500               | >600         | 300             | 300          | 250               | 600                |
| BARK: bark thickness (mm) <sup>#</sup>          | 11 (30)         | 24 (32)           | 31 (34)      | 35 (30)         | 33 (33)      | 30 (30)           | 35 (30)            |
| SPROUT: sprouting capability                    | no              | no                | no           | no              | no           | no                | yes                |
| CONE-L: cone length (cm)                        | 4.0             | 4.5               | 6.5          | 13.2            | 11.0         | 9.0               | 11.5               |
| SEED: seed weight (mg)                          | 9               | 11                | 20           | 56              | 666          | 19                | 116                |
| WING: wing length (mm)                          | 12              | 14                | 17           | 33              | 3            | 18                | 17                 |
| SEED/WING                                       | 0,75            | 0,76              | 1,14         | 1,70            | 222,00       | 1,06              | 6,82               |
| PRECOC: age of firsts flowering (years)         | >18             | 10–20             | 15–20        | 4–10            | 10–20        | 4–8               | 15–20              |
| SEROT: Serotinous cones (%)                     | 0               | 0                 | 0            | 2–82            | 0            | 40–80             | 0–60               |
| PERSIS: Cone persistence in canopy (years)      | 1–3             | 1–3               | 1–3          | 2–40            | 1–3          | 5–20              | 2–10               |
| MASTING: interval between large crops (years)*" | 3–4             | 2–4               | 2–4          | 1–2             | 3–4          | 1                 | 3–4                |
| DISPER: Dispersion date (season)*"              | Sp              | Sp                | Sp           | Sp-Su           | Sp A         | Su                | Su                 |
| VIABIL: Seed viability (years)**                | 5               | 5                 | 5            | 30              | 5            | 20                | 15                 |
| GERMI: Seed germination (%. in 1 month)         | >95             | >95               | >95          | 30–95           | >95          | >95               | 80–95              |



# FIRE ECOLOGY



Comparing bark thickness and bud tolerance to heat (bud width), Fernandes et al. (2008) ranked ***pine resistance to heat*** in the following order:

*P. canariensis* > *P. pinaster* > *P. pinea* > *P. nigra* > *P. halepensis* > *P. brutia* > *P. sylvestris* > *P. uncinata*.

However, bark thickness varies among populations, namely in *P. pinaster*.

Fire resistance rating of European pine species to crown and cambium death, based on morphological traits and lethal heating experiments. ***Pinea 1 does not accommodate the effect of canopy architecture, Pinea 2 accommodates the effect of canopy architecture.***



# Main Mediterranean Basin pines





# Management implications

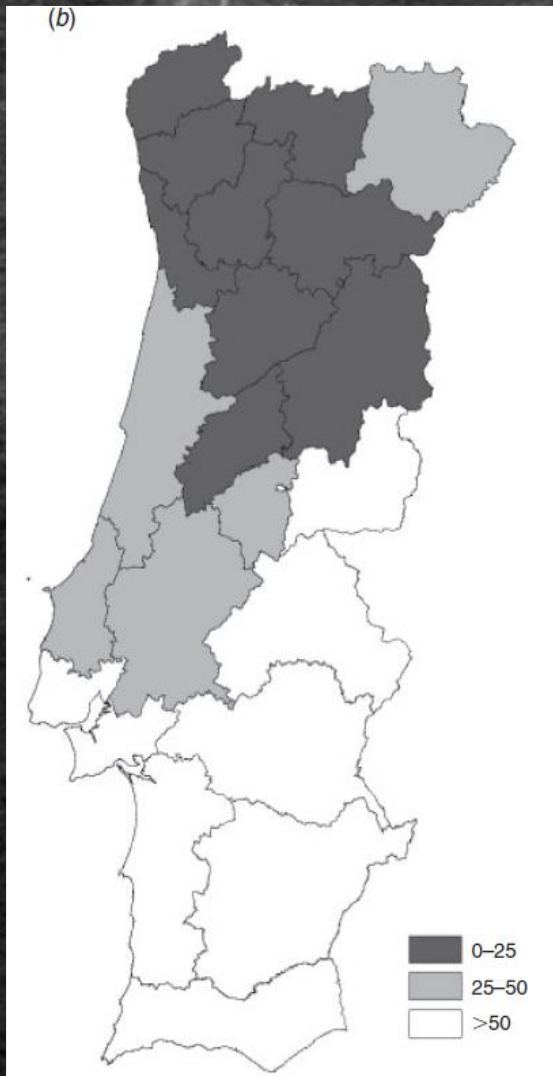
Fire ecology has practical implications both for the production of material goods and of ecosystem services.

***The most vulnerable taxa may be the postfire obligate seeders (R-S+)***, which invest their future in a single episode of postfire reproduction. This is risky because it very much depends on fire intervals long enough to allow for sufficient seed banks to develop. Therefore, obligate-seeders are sensitive to changes in fire regime, in particular ***increased fire frequency***. This applies both to tree and shrub species.

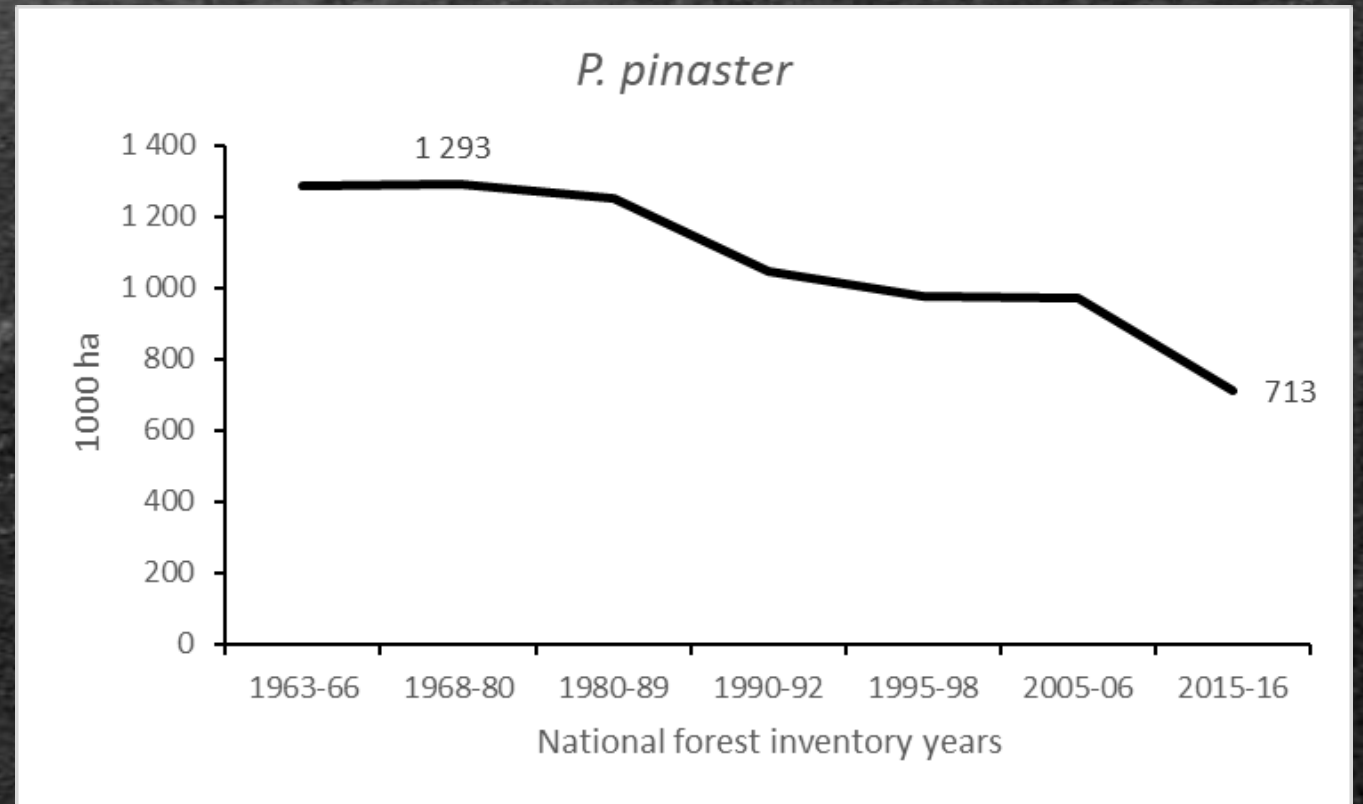
Human activity is having major impacts on fire regimes, namely on MTE, where people are the main ignition source for wildfires, and demographic and land use changes modify (shorten or extend) fire return intervals. These effects may endanger species much earlier than climate change.



# FIRE ECOLOGY



Median interval between consecutive fires by Forest Planning Region (1975 – 2005).



Between the peak area, around 1975, and the last National Forest Inventory (2015) 45% of the *P. pinaster* area was lost. This period of 40 years corresponds to the time of a single rotation for the species. **Immaturity risk**, i.e. the reoccurrence of fire before an obligate seeder reaches sexual maturity **is a significant driver of this trend**.



# Management implications

Afforestation and postfire restoration projects typically use **local seed sources from good quality stands to maximize plant establishment and growth**. These local stands are selected on the basis of bioclimatic and geographic conditions and on high seed production. However, there is growing evidence that these seed sources are not always the best, especially under changing conditions or in highly fragmented landscapes.

Evolutionary fire ecology suggests that afforestation of pines in ecosystems susceptible to fire would be more sustainable if done with **seeds of populations with the appropriate traits to persist under the expected fire regimes**. Since **serotiny is heritable**, the postfire generation of a fire-embracer pine would be more serotinous than a mature population in a good-quality site and, thus, this seed source would be more suited for restoring an area prone to crown fires.

By selecting seeds from young postfire populations, we are also selecting for individuals with **early maturation**, which is also adaptive in ecosystems with crown fires.

Fire ecological knowledge will be more important under **closer-to-nature silvicultural practices**, namely stronger reliance on **assisted natural regeneration** of burned pine forests.



# Management implications

Considering the main fire syndromes in Mediterranean Basin pines, we can interpret preventive silviculture interventions like clearing the forest underbrush and pruning the trees, e.g. in pine and eucalypt plantations, as **attempts to emulate the structure of fire-tolerant forests**, while relying on fire-embracer tree species.

This is most evident with ***P.pinaster***, which is considered **fire tolerant in natural stands** in the coastal western Mediterranean Basin but, as currently planted, the industrial forestry stands **end up behaving as those of natural fire-embracer species**.

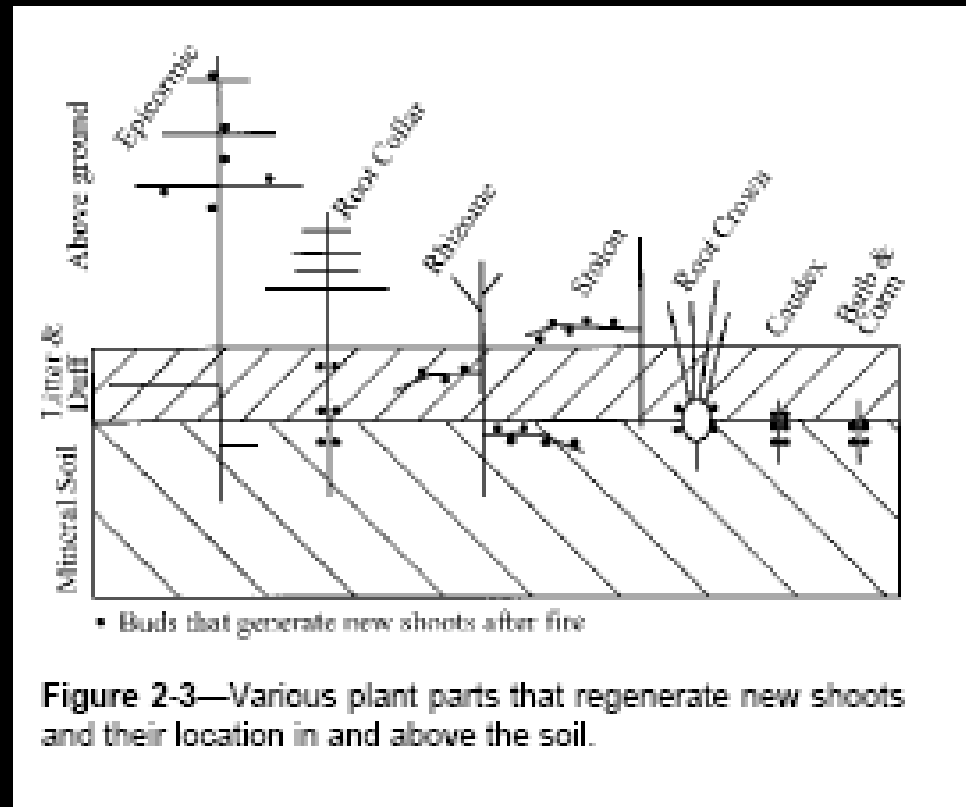
The combination of increased stand density and under-management (insufficient underbrush clearing, standing thinning, and pruning) brings along the corresponding intensity of fire behavior and severity of effects characteristic of fire embracers.

## Shrub fuel management



# ECOLOGIA DO FOGO

## PROTECÇÃO DE TECIDOS CRÍTICOS – Protecção vegetativa





# ECOLOGIA DO FOGO

## ADAPTAÇÕES DOS PINHEIROS AO FOGO





# ECOLOGIA DO FOGO

## ADAPTAÇÕES DOS PINHEIROS AO FOGO

*P. canariensis*



*P. pinea*





# ECOLOGIA DO FOGO

## ADAPTAÇÕES DOS PINHEIROS AO FOGO





# ECOLOGIA DO FOGO

## ADAPTAÇÕES DOS PINHEIROS AO FOGO





# FIRE ECOLOGY

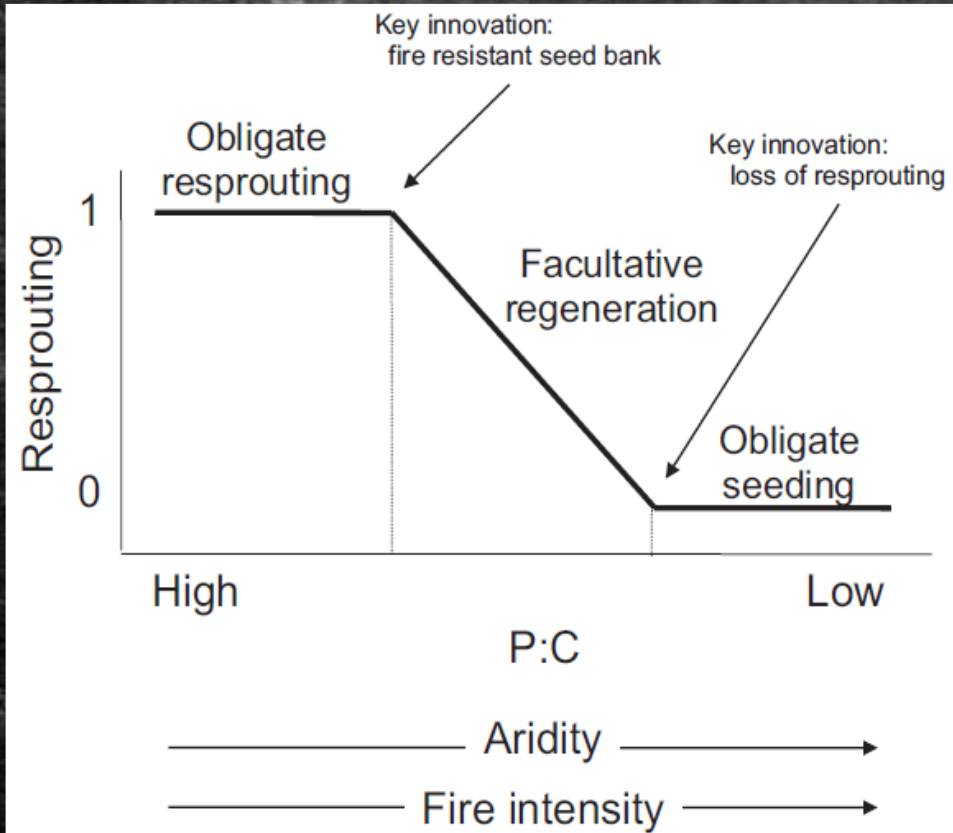


Fig. 3 Model of Fig. 1 applied to fire-prone ecosystems. Changes in the probability of resprouting along an adult-to-offspring survival ratio ( $P:C$ ) gradient are not linear but show two turning points related to the acquisition of key innovations: the capacity to store a fire-resistant seed bank (postfire seeding), and the loss of resprouting capacity. Changes in the  $P:C$  ratio may be produced by different drivers (Fig. 2) and may have driven the rise of innovations during evolution.

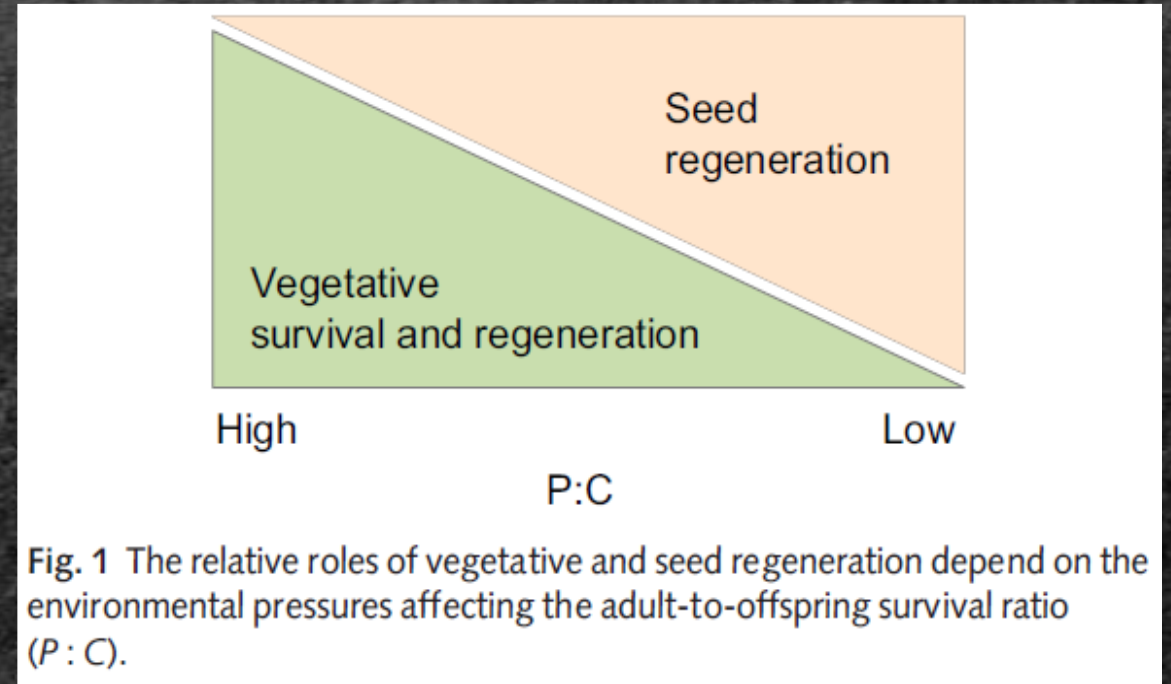


Fig. 1 The relative roles of vegetative and seed regeneration depend on the environmental pressures affecting the adult-to-offspring survival ratio ( $P:C$ ).



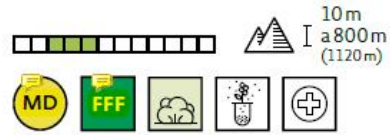
# FIRE ECOLOGY



**Cistus ladanifer**  
subsp. *ladanifer*

ESTEVA, XARA

**Ecologia:** matos; em solos secos e pobres, ácidos.



**Calluna vulgaris**

TORGA, QUEIRÓ, URZE

**Ecologia:** matos acidófilos, pinhais, desde dunas até alta montanha.



**Erica australis**

URGUEIRA, URZE-VERMELHA, TORGA-VERMELHA

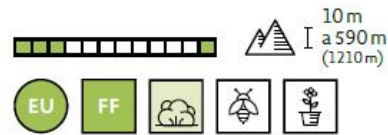
**Ecologia:** matos acidófilos; em locais soalheiros.



**Erica lusitana**

URZE-PORTUGUESA

**Ecologia:** matos higrófilos; em solos ácidos.



**Erica arborea**

URZE-BRANCA, URZE-ARBÓREA

**Ecologia:** matagais; em encostas frescas.



**Cytisus multiflorus**

GIESTA-BRANCA

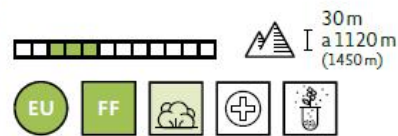
**Ecologia:** matos, pastagens; em solos ácidos e pobres.



**Cytisus scoparius**

GIESTA-AMARELA, GIESTEIRA-DAS-VASSOURAS, CHAMIÇA

**Ecologia:** matos, orlas florestais, sebes, bermas de caminhos; em solos ácidos.



**Genista triacanthos**

TOJO-GATANHO-MENOR, ARRANHA-LOBO

**Ecologia:** matos, bosques; em solos ácidos.



**Pterospartum tridentatum**

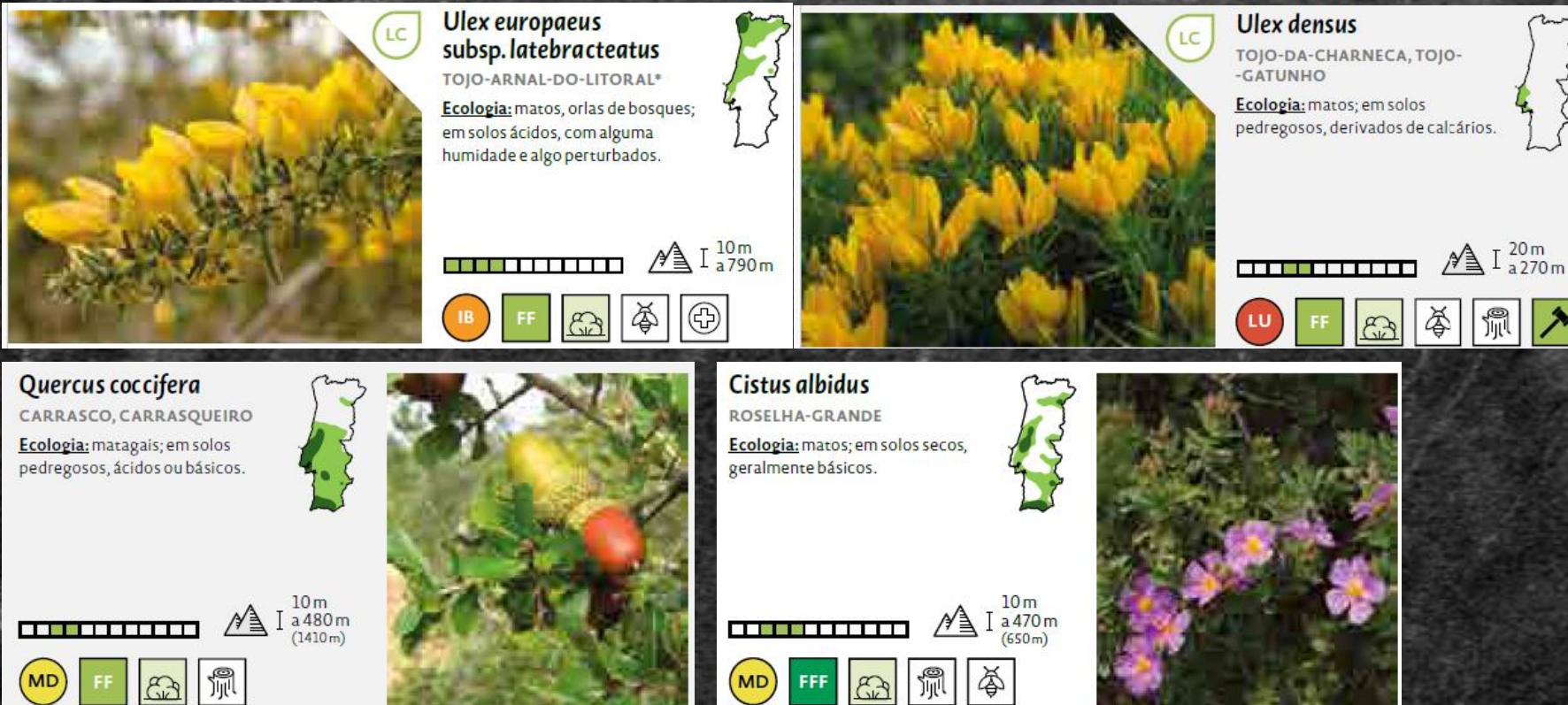
CARQUEJA, CARQUEIJA

**Ecologia:** matos; em solos ácidos; do litoral à alta montanha.



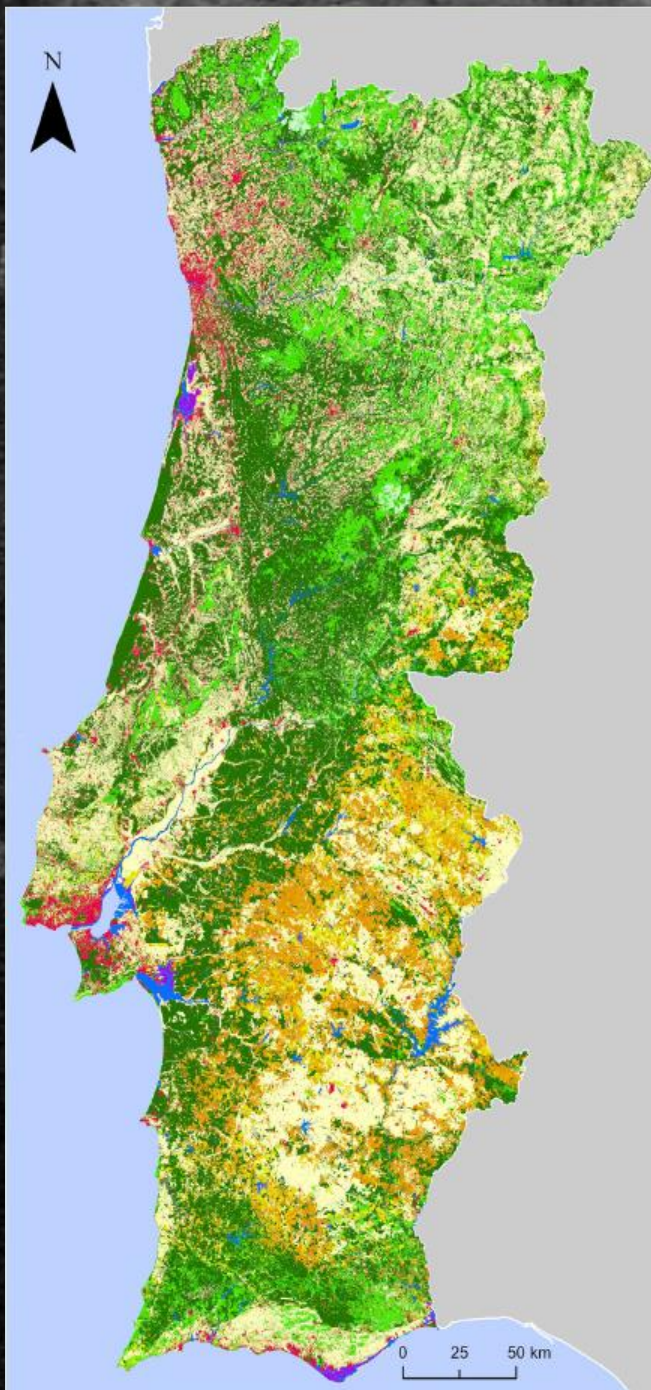


# FIRE ECOLOGY

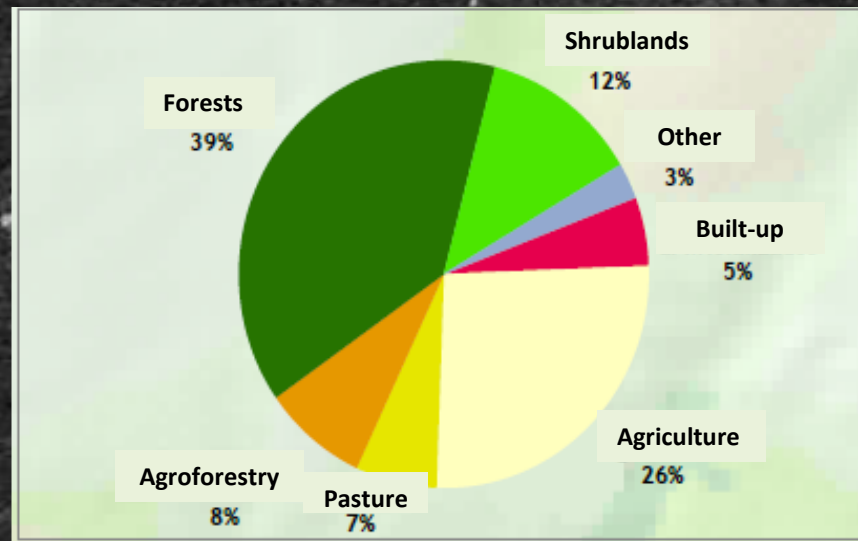




# FIRE ECOLOGY



Major land cover classes in Portugal (COS 2018)





# FIRE ECOLOGY

| Taxon                           | Tree | SLA  | N <sub>2</sub> -fixing capacity | Resprouting ability | Reproductive strategy             |
|---------------------------------|------|------|---------------------------------|---------------------|-----------------------------------|
| <i>Eucalyptus globulus</i>      | Yes  | Low  | No                              | Yes                 | R <sub>aboveground</sub>          |
| <i>Pinus pinaster</i>           | Yes  | Low  | No                              | No                  | S <sub>heavy, no-stimulated</sub> |
| <i>Quercus suber</i>            | Yes  | Low  | No                              | Yes                 | R <sub>aboveground</sub>          |
| <i>Calluna vulgaris</i>         | No   | Low  | No                              | No                  | S <sub>light, no-stimulated</sub> |
| <i>Cistus albidus</i>           | No   | Low  | No                              | No                  | S <sub>light, stimulated</sub>    |
| <i>Cistus ladanifer</i>         | No   | Low  | No                              | No                  | S <sub>light, stimulated</sub>    |
| <i>Cistus monspeliensis</i>     | No   | Low  | No                              | No                  | S <sub>light, stimulated</sub>    |
| <i>Cistus salviifolius</i>      | No   | Low  | No                              | No                  | S <sub>light, stimulated</sub>    |
| <i>Erica umbellata</i>          | No   | High | No                              | No                  | S <sub>light, no-stimulated</sub> |
| <i>Rosmarinus officinalis</i>   | No   | Low  | No                              | No                  | S <sub>light, no-stimulated</sub> |
| <i>Ulex parviflorus</i>         | No   | Low  | Yes                             | No                  | S <sub>heavy, stimulated</sub>    |
| <i>Erica arborea</i>            | No   | High | No                              | Yes                 | R <sub>aboveground</sub>          |
| <i>Pterospartum tridentatum</i> | No   | Low  | Yes                             | Yes                 | R <sub>aboveground</sub>          |
| <i>Erica australis</i>          | No   | High | No                              | Yes                 | R <sub>underground</sub>          |
| <i>Erica multiflora</i>         | No   | Low  | No                              | Yes                 | R <sub>underground</sub>          |
| <i>Pistacia lentiscus</i>       | No   | Low  | No                              | Yes                 | R <sub>underground</sub>          |
| <i>Quercus coccifera</i>        | No   | Low  | No                              | Yes                 | R <sub>underground</sub>          |
| <i>Genista florida</i>          | No   | -    | Yes                             | Yes                 | F                                 |
| <i>Retama sphaerocarpa</i>      | No   | -    | Yes                             | Yes                 | F <sub>heavy</sub>                |
| <i>Genista scorpius</i>         | No   | Low  | Yes                             | Yes                 | F <sub>heavy, no-stimulated</sub> |
| <i>Cytisus scoparius</i>        | No   | High | Yes                             | Yes                 | F <sub>heavy, stimulated</sub>    |
| <i>Ulex europaeus</i>           | No   | High | Yes                             | Yes                 | F <sub>heavy, stimulated</sub>    |
| <i>Erica cinerea</i>            | No   | Low  | No                              | Yes                 | F <sub>light, no-stimulated</sub> |
| <i>Thymus vulgaris</i>          | No   | Low  | No                              | Yes                 | F <sub>light, no-stimulated</sub> |

Fire-related functional traits of Mediterranean-type shrub species and of the three most widespread forest species in Portugal.

**R** = obligate resprouter

**S** = obligate seeder

**F** = facultative, i.e. both strategies

**aboveground** = resprouters and facultatives without underground buds

**underground** = resprouters and facultatives with underground buds

**light** = obligate seeders and facultatives with light seeds

**heavy** = obligate seeders and facultatives with heavy seeds

**stimulated** = obligate seeders and facultatives with heat-stimulated germination

**no-stimulated** = obligate seeders and facultatives without heat-stimulated germination

**SLA** = Specific leaf area



# FIRE ECOLOGY

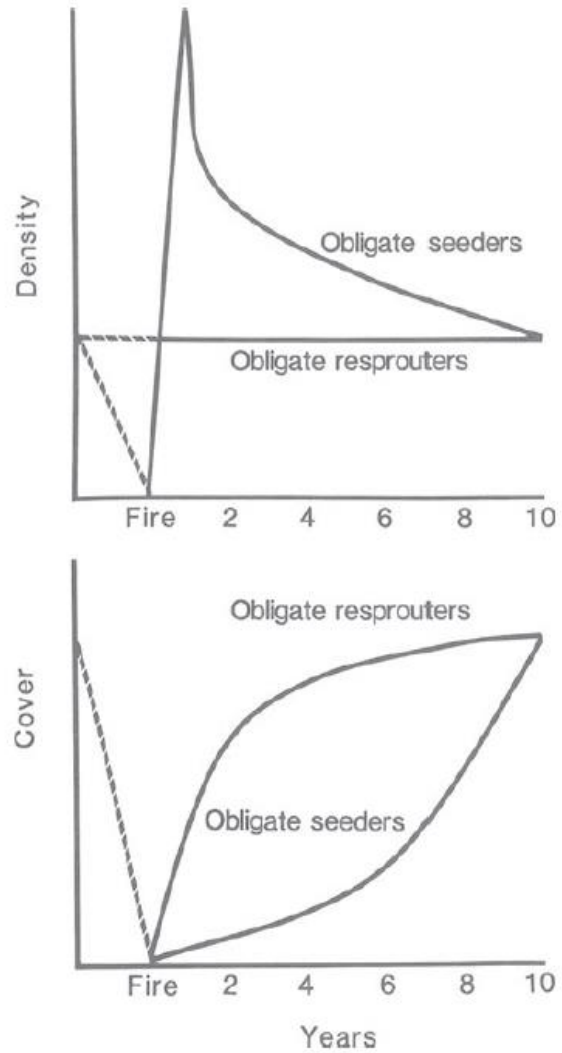


Fig. 2. Schematic description of demographic changes in obligate resprouters and obligate seeders in the immediate postfire environment.

Table 1. Generalized mechanisms of species response to fire (strategy), their fire dynamics and persistence scale, the prevalence for animals and plants in fire prone ecosystems (low, moderate and high) and some traits related to strategy. The last column refers to the fire characteristics where this strategy is most likely to occur ('high' and 'low' refers to fire intensity; 'patchy' refers to high level of patchiness; 'seasonal' refers to strong seasonality; 'large' refers to the size of the area burnt); and more details can be found in the text. Individual persistence implies low postfire mortality while the other persistence scales implies high postfire mortality. Some examples of each strategy are provided in the text.

| Strategy               | Dynamics           | Persistence scale | Kingdom           | Prevalence  | Traits                                                                                | Fire regime         |
|------------------------|--------------------|-------------------|-------------------|-------------|---------------------------------------------------------------------------------------|---------------------|
| Resistance             | fire survival      | individual        | plants<br>animals | high<br>low | resprouting, thick bark<br>thick cover, physiological<br>tolerance                    | any<br>low          |
| Refugia                | fire survival      | individual        | plants<br>animals | low<br>high | lack of fire related traits<br>fire detectors, ability to quickly<br>react (hide)     | patchy<br>patchy    |
| Avoidance              | fire survival      | individual        | plants<br>animals | mod<br>low  | low flammability, self-pruning<br>behavioral traits to reduce<br>habitat flammability | low<br>low          |
| Crypsis                | postfire survival  | individual        | plants<br>animals | low<br>mod  | cryptic seeds<br>cryptic body, torpor                                                 |                     |
| Dormant                | propagule survival | population        | plants<br>animals | high<br>low | seed dormancy, serotiny<br>underground life stages during<br>the fire season          | high<br>seasonal    |
| Exogenous recolonizing | postfire dynamics  | landscape         | plants<br>animals | low<br>high | high dispersal (wind)<br>high mobility (large, with<br>wings)                         | high<br>high, large |
| Intolerance            | local extinction   | region            | plants<br>animals | no<br>no    | lack of fire related traits<br>lack of fire related traits                            | no<br>no            |



# FIRE ECOLOGY

**Succession**: the changes displayed by an ecosystem over time following a disturbance. These changes occur in a range of ecosystem attributes, such as species diversity and composition, vegetation structure, ecosystem processes and soils.

**Successional pathway**: temporal change in state variables displayed by a successional system. It can encompass changes in community types, species composition and abundance.

**Successional mechanism**: a factor (e.g. seed availability) or process (e.g. facilitation, competition) that causes successional change. Mechanisms can operate at different organizational levels, such as the species level or community level.

**Successional model**: a conceptual representation of how successional change works. It specifies the relationship between the various stages of the successional pathway and the successional mechanisms that cause the change from one stage to the next. Such a model can be verbal, diagrammatic, or quantitative.



# FIRE ECOLOGY

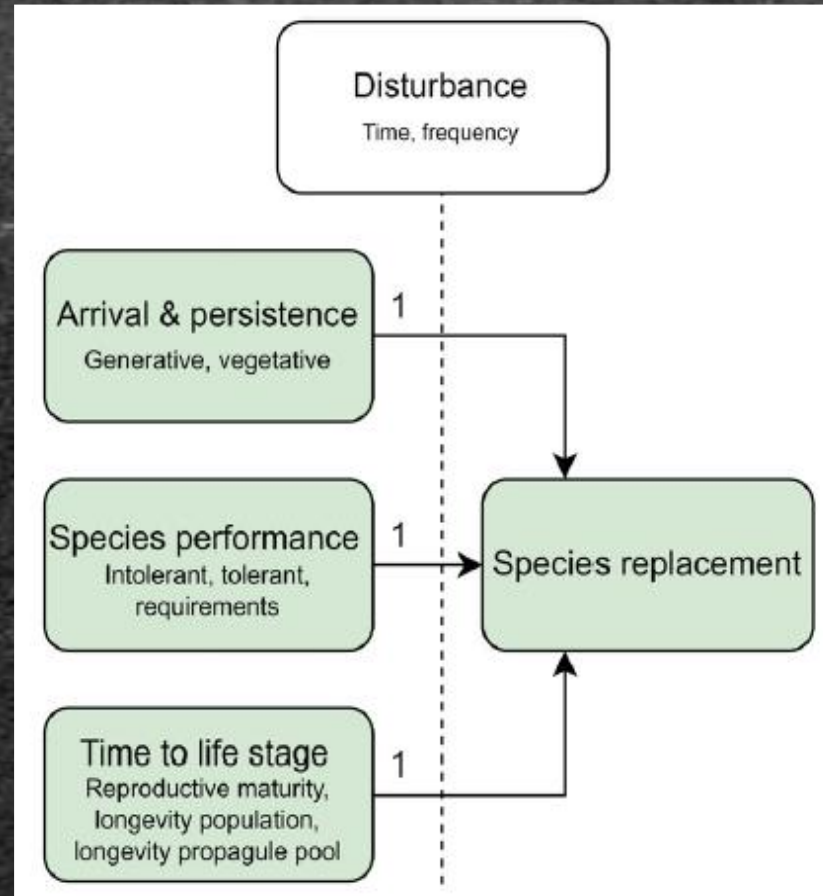


Fig. 11. Vital attributes (Noble & Slatyer, 1980). Changes in species composition during succession are determined by the disturbance regime (timing, frequency) that filters out species (as indicated by the vertical dashed line) based on their vital attributes (1). Three groups of vital attributes are distinguished, related to arrival and persistence (generative or vegetative regeneration), species performance (i.e. the ability to establish and grow in the community: intolerant, tolerant, and specific later successional requirements), and the time to reach a life stage (reproductive maturity, longevity of species population in the community, longevity of the propagule pool).



# FIRE ECOLOGY

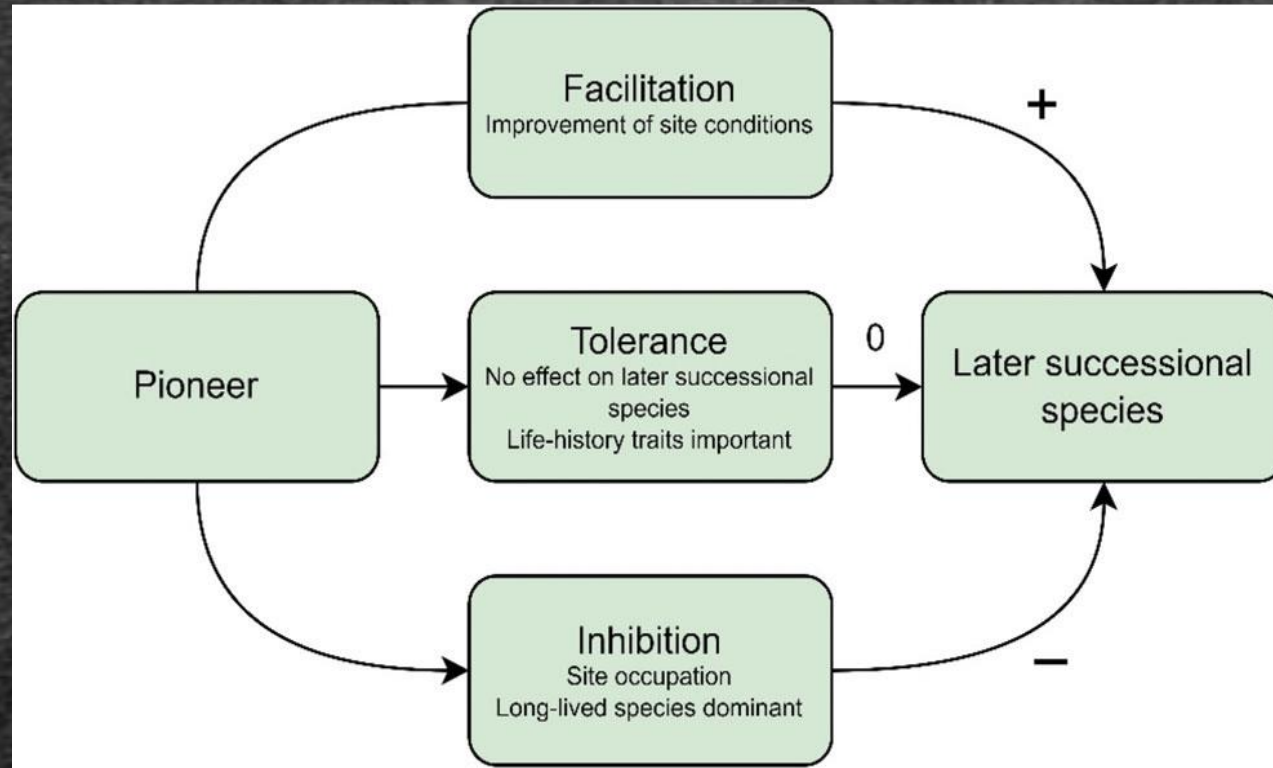


Fig. 9. Facilitation, tolerance, and inhibition (Connell & Slatyer, 1977). Pioneers can affect succession by later successional species in three ways. Pioneers can have a positive effect (facilitation) by improving site conditions, a neutral effect (tolerance) because later successional species are tolerant to changed environmental conditions, or a negative effect (inhibition) by occupying the site and preventing the establishment of other species.



# FIRE ECOLOGY

| No                                | Trait            | Trait full name                         | Data types                                                   | Trait type |
|-----------------------------------|------------------|-----------------------------------------|--------------------------------------------------------------|------------|
| <b>General vegetative traits</b>  |                  |                                         |                                                              |            |
| 1                                 | GrowthForm       | Growth form                             | categorical                                                  | F          |
| 2                                 | LeafDivision     | Leaf division degree                    | categorical                                                  | F          |
| 3                                 | LeafLifespan     | Average leaf lifespan                   | quantitative                                                 | V          |
| 4                                 | LeafPhenology    | Leaf phenology                          | categorical                                                  | F          |
| 5                                 | LeafShape        | Basic leaf shape                        | categorical                                                  | F          |
| 6                                 | LeafArea         | Average leaf area                       | quantitative or semi-quantitative                            | V          |
| 7                                 | LNCn             | Mass-based leaf nitrogen content        | quantitative                                                 | V          |
| 8                                 | SLA              | Average specific leaf area              | quantitative                                                 | V          |
| 9                                 | BarkThick        | Average bark thickness                  | quantitative, quantitative conditional, or semi-quantitative | V          |
| 10                                | Height           | Average height                          | quantitative                                                 | V          |
| 11                                | StemDensity      | Stem specific density                   | quantitative                                                 | V          |
| 12                                | CFFuel           | Coarse fine fuel                        | quantitative or semi-quantitative                            | V          |
| 13                                | DeadFuel         | Dead fine fuel                          | quantitative or semi-quantitative                            | V          |
| 14                                | LDMC             | Leaf dry matter content                 | quantitative                                                 | V          |
| 15                                | Clonality        | Clonality                               | categorical or Boolean                                       | V          |
| 16                                | d13C             | Integrated water-use efficiency         | quantitative                                                 | V          |
| 17                                | Lifespan         | Lifespan                                | quantitative or semi-quantitative                            | V          |
| 18                                | NutritionalRelat | Nutritional relationships               | categorical                                                  | V          |
| 19                                | P50              | Resistance to xylem cavitation          | quantitative                                                 | V          |
| 20                                | RootDepth        | Maximum rooting depth                   | quantitative                                                 | V          |
| 21                                | Spinescence      | Spinescence                             | Boolean                                                      | F          |
| 22                                | SRR              | Shoot:root ratio                        | quantitative or semi-quantitative                            | V          |
| <b>Sexual reproductive traits</b> |                  |                                         |                                                              |            |
| 23                                | DispMode         | Dispersal mode                          | categorical                                                  | F          |
| 24                                | Diaspore         | Diaspore                                | categorical                                                  | F          |
| 25                                | FruitMass        | Fruit dry mass                          | quantitative or semi-quantitative                            | V          |
| 26                                | FruitType        | Fruit type                              | categorical                                                  | F          |
| 27                                | SeedMass         | Seed dry mass                           | quantitative or semi-quantitative                            | V          |
| 28                                | SeedProd         | Annual seed production                  | quantitative or semi-quantitative                            | V          |
| 29                                | SeedShape        | Basic seed shape                        | quantitative or semi-quantitative                            | V          |
| <b>Regeneration traits</b>        |                  |                                         |                                                              |            |
| 30                                | BadSource        | Bad source                              | categorical                                                  | V          |
| 31                                | FireStimFlower   | Fire-stimulated flowering               | semi-quantitative or Boolean                                 | V          |
| 32                                | RespClip         | Resprouting capacity after clipping     | quantitative, semi-quantitative, categorical or Boolean      | V          |
| 33                                | RespDist         | Resprouting capacity after disturbances | quantitative, semi-quantitative, categorical or Boolean      | V          |
| 34                                | RespFire         | Resprouting capacity after Fire         | quantitative, semi-quantitative, categorical or Boolean      | V          |
| 35                                | ChemCues         | Chemical germination cues               | semi-quantitative conditional                                | V          |
| 36                                | HeatStimGerm     | Heat-stimulated germination             | semi-quantitative conditional or Boolean conditional         | V          |
| 37                                | OtherCues        | Other germination cues                  | semi-quantitative conditional                                | V          |
| 38                                | CanopySeedBank   | Canopy seed bank longevity              | semi-quantitative or Boolean                                 | V          |
| 39                                | SerotinyLevel    | Serotiny level                          | quantitative or quantitative range                           | V          |
| 40                                | SoilSeedBank     | Soil seed bank longevity                | semi-quantitative conditional                                | V          |
| 41                                | MatResp          | Age at maturity of resprouts            | quantitative or semi-quantitative                            | V          |
| 42                                | MatSap           | Age at maturity of saplings             | quantitative or semi-quantitative                            | V          |
| 43                                | SeedEmerg        | Post-fire seedling emergence            | quantitative, semi-quantitative, categorical or Boolean      | V          |
| 44                                | SeedSurv         | Post-fire seedling survival             | quantitative or semi-quantitative                            | V          |

Table 9. List of functional traits included in the database. ‘Trait’ refers to the short name of functional trait as used in the database, ‘Trait full name’ is the full name of functional traits, ‘Data types’ shows the types of data used in each functional trait, and ‘Trait type’ shows whether a functional trait is fixed (F, i.e. one record for each species) or variable (V, any species may include several records reflecting variability of the trait).



# FIRE ECOLOGY

## A SUCESSÃO ECOLÓGICA E O FOGO

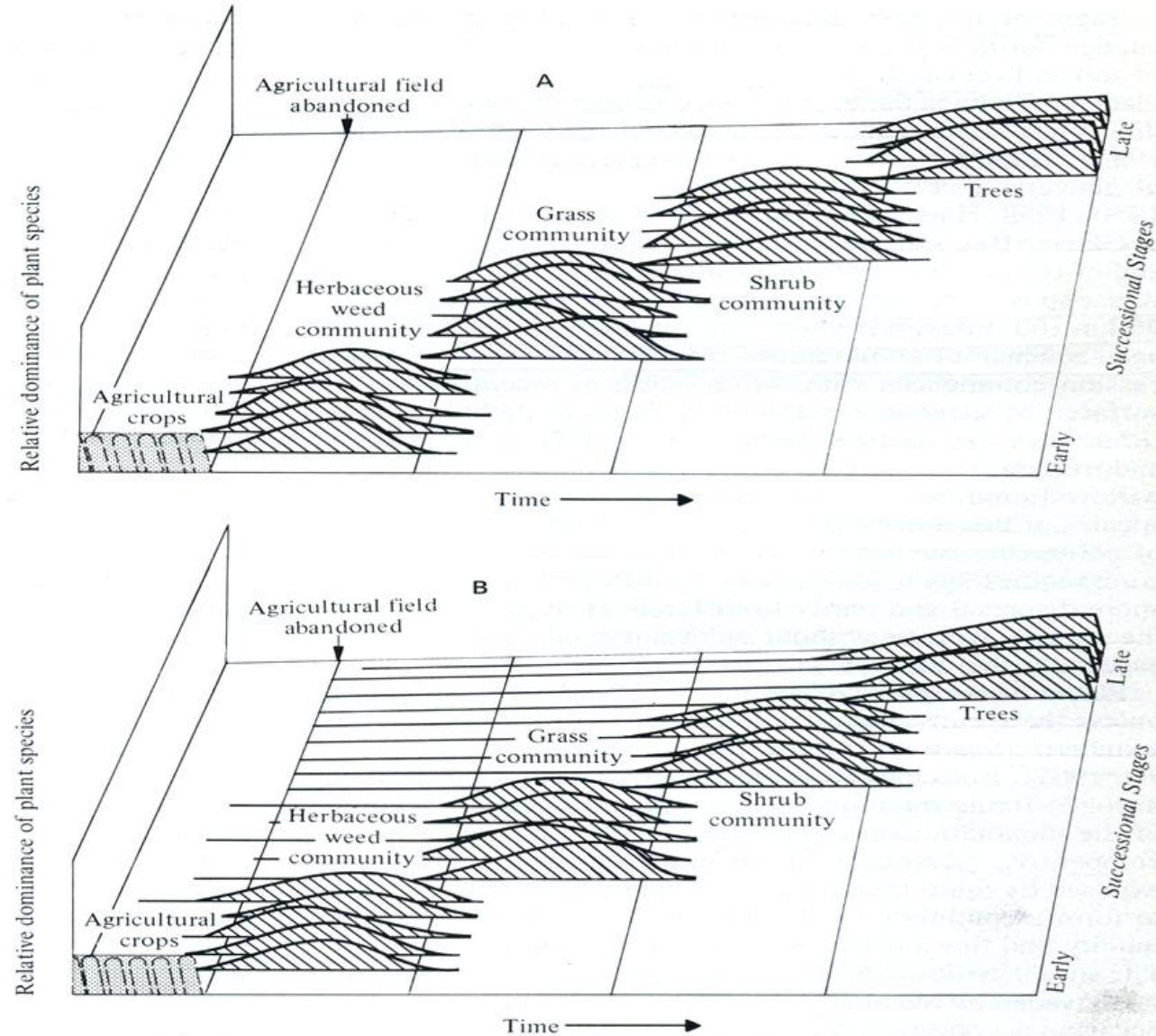
- Um aspecto fundamental da sucessão pós-fogo é a re-colonização (Egler, 1954) da área queimada.
- A perspectiva tradicional da colonização considerava vagas sucessivas de plantas, chegando e estabelecendo-se com sucesso numa dada área disponível e sendo substituídas pela vaga seguinte de novas colonizadoras (*relay floristics RF*, ou colonização “por estafetas”). Cada estágio da sucessão prepararia o terreno para o estágio seguinte.
- Porém, nalguns tipos de sucessão, é óbvio que há propágulos de plantas características de fases avançadas da sucessão, que estão presentes desde o início. As diferentes fases da sucessão representam apenas os períodos variáveis de ocupação antes da espécie, ou grupo de espécies, se tornar dominante (“composição florística inicial, CFI”).



# FIRE ECOLOGY

**Figure 15.4**

Diagrammatic representation of (A) relay floristics and (B) initial floristic composition types of old-field succession as envisaged by Egler (1954). Most successions will exhibit some combination of these two patterns. (Reproduced by permission of the International Society for Vegetation Science and the author.)





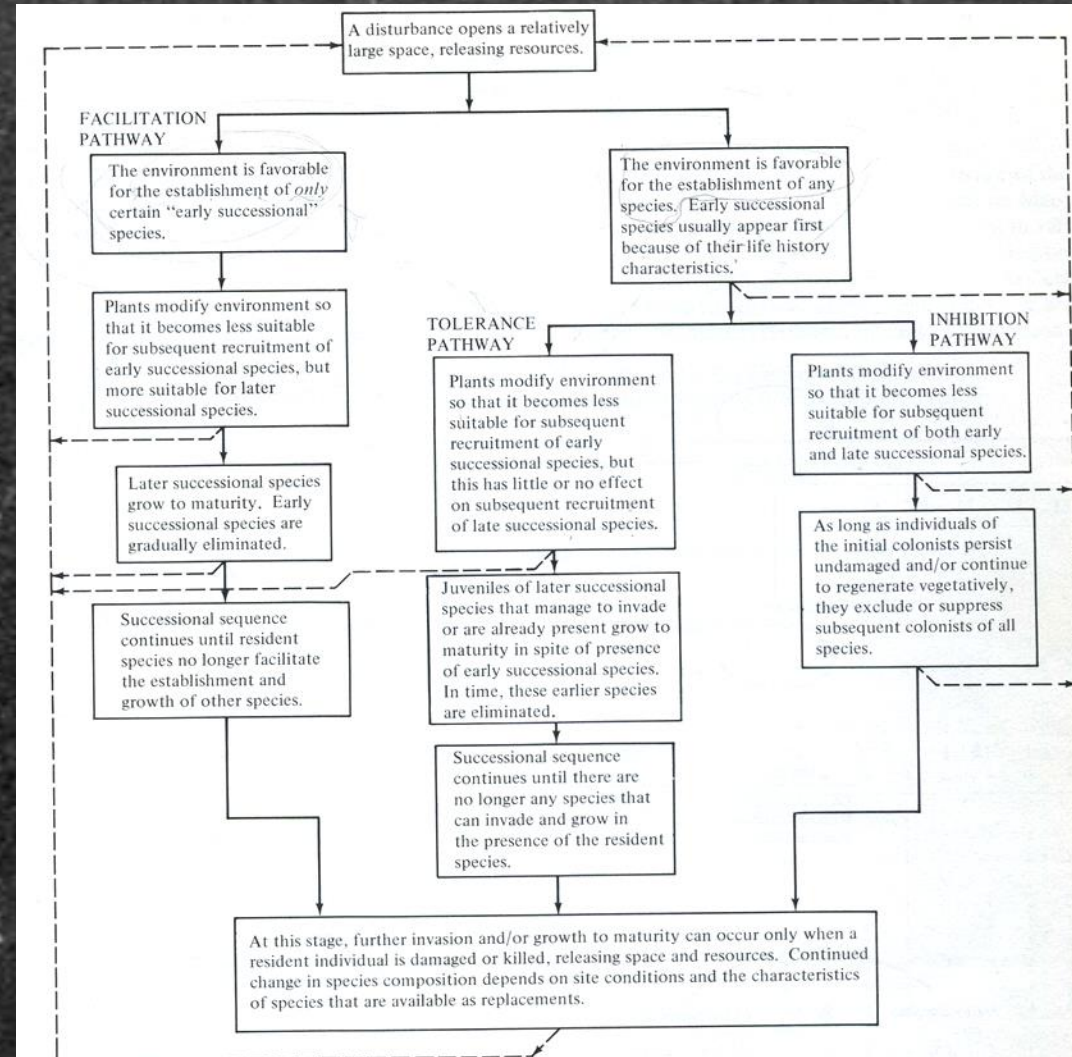
# FIRE ECOLOGY

## A SUCESSÃO ECOLÓGICA E O FOGO

- Muitos dos modelos tradicionais de sucessão ecológica consideram um único padrão de substituição successional da vegetação.
- Connell e Slatyer (1977) propuseram um modelo mais complexo, incluindo **3 vias** sucessionais:
  - via da **facilitação**, equivalente ao modelo RF de Egler.
  - via da **tolerância**, relacionada com a CFI: o ambiente, no início do processo, permite a invasão por qualquer tipo de espécie. As primeiras a surgir são as adaptadas à colonização rápida de locais afectados por perturbações e que são intolerantes ao ensombramento. Acabam excluídas competitivamente pelas espécies tolerantes ao ensombramento, típicas das fases mais avançadas da sucessão.
  - via da **inibição**. As espécies ocupantes iniciais de um local inibem o estabelecimento de propágulos de outras espécies, quer das fases iniciais, quer das fases avançadas da sucessão. A substituição de espécies ocorre apenas quando a morte de indivíduos ou uma perturbação criam novas possibilidades de colonização.



# FIRE ECOLOGY



**Figure 15.11**

The three-pathway model of succession of Connell and Slatyer, 1977. (Copyright The University of Chicago Press. Used with permission of the publisher and authors.)



# FIRE ECOLOGY

## SUCESSÃO ECOLÓGICA E FOGO

- Embora o modelo das 3-vias represente um grande progresso relativamente aos modelos clássicos, foi sujeito a críticas, por não ser aplicável a ecossistemas dominados pelo fogo, ser incapaz de quantificar os efeitos de distúrbios de diferentes intensidades e frequências, e por não incorporar explicitamente adaptações das espécies que modulam as suas respostas a diferentes tipos de distúrbio.
- Em alternativa, foi proposto um modelo de **vias múltiplas** para a sucessão, que requer três tipos de informação para cada espécie com um papel dominante na sucessão:
  - método de persistência
  - condições para o estabelecimento
  - eventos críticos da história vital



# FIRE ECOLOGY

## SUCESSÃO ECOLÓGICA E FOGO

- **Método de persistência:** o método de invasão e colonização, ou persistência dos propágulos após um distúrbio ou perturbação (4 classes).
  - persistência por chegada de sementes de fonte remota (**espécies D:** sementes dispersadas).
  - persistência por sementes de longa viabilidade armazenadas no solo. Este banco pode persistir através de vários distúrbios sem reabastecimento, porque nem todas as sementes germinam após a 1ª perturbação (**espécies S:** banco no solo).
  - persistência de sementes em pinhas ou frutos protetores situados na copa; sementes apenas disponíveis se houver plantas sexualmente maduras antes do distúrbio (**espécies C:** banco na copa).
  - persistência via órgão vegetativo protegido que rebrota prontamente após o distúrbio (**espécies V:** regeneração vegetativa).



# FIRE ECOLOGY

## SUCESSÃO ECOLÓGICA E FOGO

- **Condições para estabelecimento:** as condições requeridas, ou toleradas para o estabelecimento e crescimento até à maturidade (3 classes).
  - espécies capazes de se estabelecer em qualquer momento, na presença de indivíduos maduros da mesma ou de outras espécies; capazes de tolerar a competição (**espécies T**: tolerantes).
  - espécies capazes de se estabelecer apenas imediatamente após o distúrbio, quando a competição é reduzida; intolerantes da competição (**espécies I**: intolerantes).
  - espécies que só se estabelecem após o ambiente ter sido modificado pela presença de indivíduos maduros da mesma ou de outras espécies (**espécies R**: ???)



# FIRE ECOLOGY

## SUCESSÃO ECOLÓGICA E FOGO

■ **Eventos críticos da história vital:** o tempo necessário para atingir estágios críticos na história vital, como idade reprodutiva ou longevidade (4 classes).

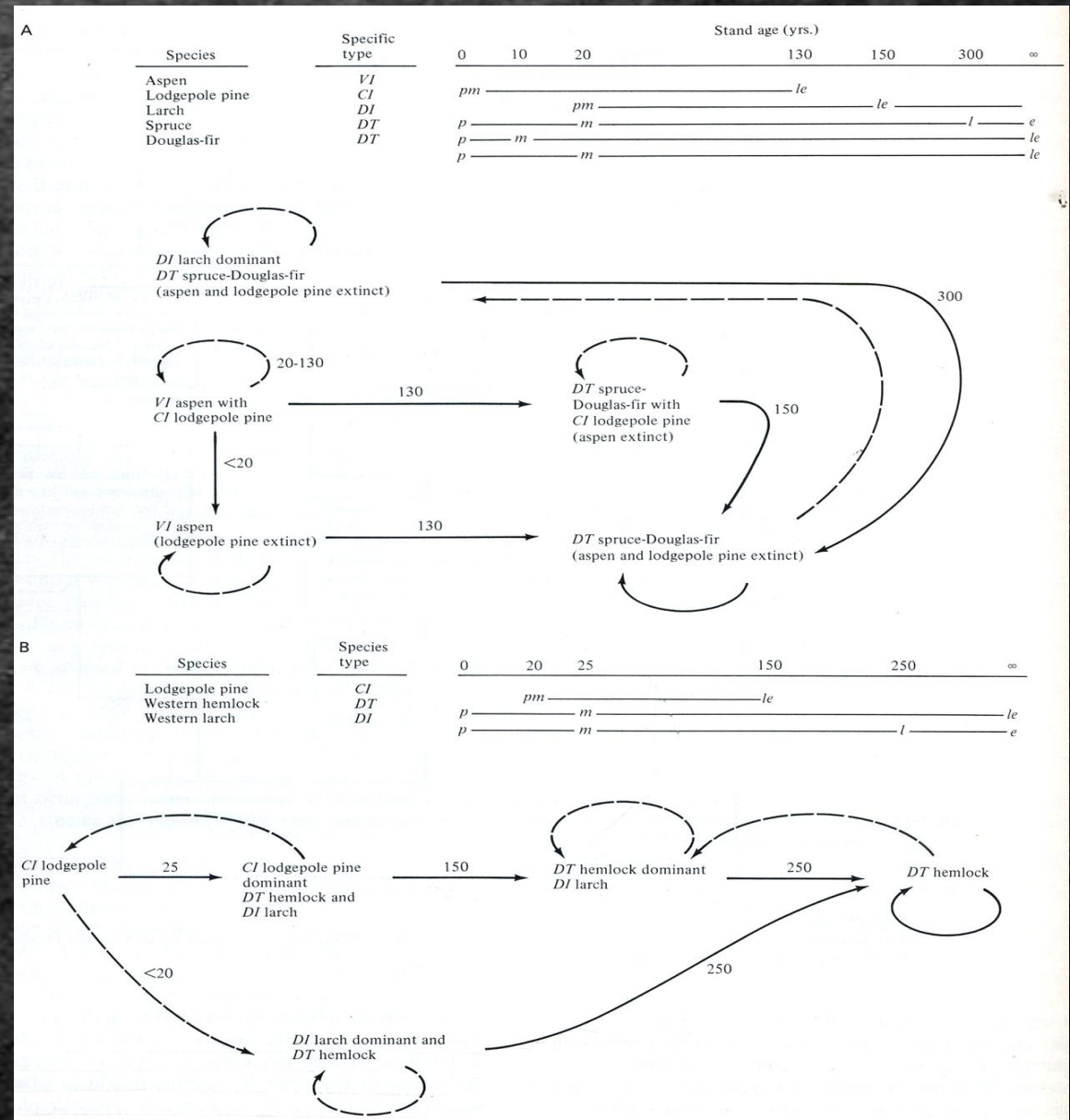
- reabastecimento de propágulos suficiente para repôr a espécie após outro distúrbio (**evento p**).
- chegada à maturidade: produção de propágulos suficientes para permitir à espécie persistir a outro distúrbio (**evento m**)
- senescência e perda da espécie da comunidade (**evento l**).
- perda de propágulos e eliminação da espécie (**evento e**).



# FIRE ECOLOGY

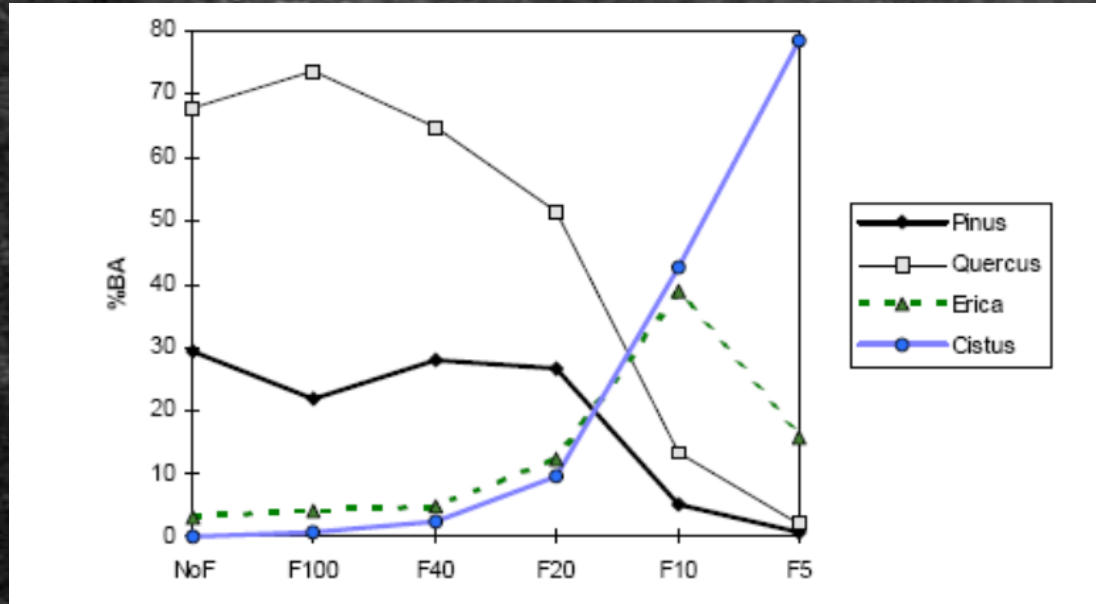
## SUCESSÃO ECOLÓGICA E FOGO

■ Usando esta informação, podem fazer-se diagramas ou modelos que prevêm e sumarizam a sucessão secundária, considerando aspetos como a frequência e intensidade do fogo, a altura das chamas e a queima das copas, no processo de substituição sucessional das espécies.





# FIRE ECOLOGY



Summary of the BROLLA simulation results for 6 scenarios with four species groups (named *Quercus*, *Pinus*, *Erica*, and *Cistus*). The scenarios are: No fire (NF) and fire every 100 (F100), 40 (F40), 20 (F20), 10 (F10), and 5 (F5) years. The y-axis represents the mean percent basal area of 10 replicated simulations for 500 years.

